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NEUROSCIENCE



HOW TO MEASURE AND REDUCE THE CARBON FOOTPRINT OF FMRI DATA ANALYSIS

NICK SOUTER

Volume 1
2023



Imaging
Neuroscience

December 14 2023

Ten recommendations for reducing the carbon footprint of research computing in human neuroimaging

Nicholas E. Souter , Loïc Lannelongue, Gabrielle Samuel, Chris Racey, Lincoln J. Colling, Nikhil Bhagwat, Raghavendra Selvan, Charlotte L. Rae



> Author and Article Information

Imaging Neuroscience (2023) 1: 1–15.

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HUMAN BRAIN MAPPING

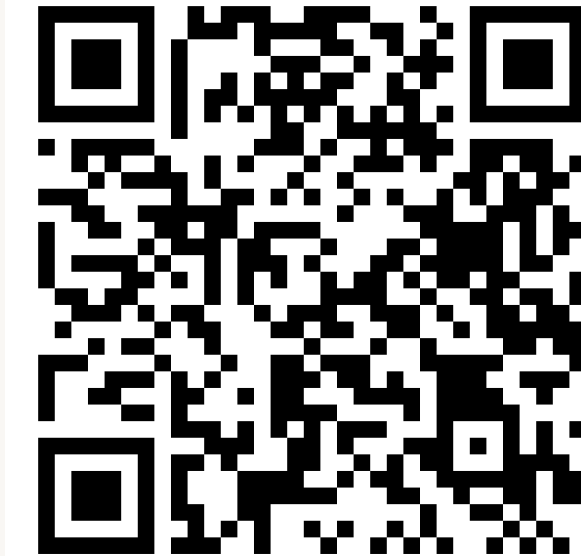


TECHNICAL REPORT |  Open Access |  

Measuring and reducing the carbon footprint of fMRI preprocessing in fMRIPrep

Nicholas E. Souter , Nikhil Bhagwat, Chris Racey, Reese Wilkinson, Niall W. Duncan, Gabrielle Samuel, Loïc Lannelongue, Raghavendra Selvan, Charlotte L. Rae

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Volume 3
2025

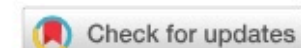


Imaging
Neuroscience

June 16 2025

Comparing the carbon footprint of fMRI data processing and analysis approaches

Nicholas E. Souter ,  Chris Racey , Nikhil Bhagwat , Reese Wilkinson , Niall W. Duncan , Gabrielle Samuel , Loïc Lannelongue , Raghavendra Selvan , Charlotte L. Rae 

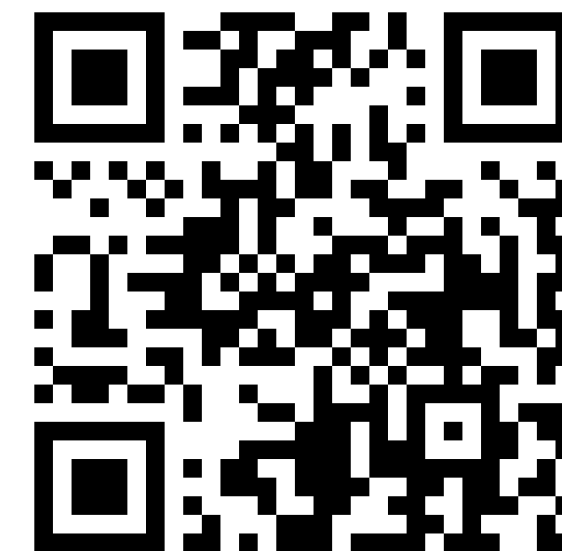


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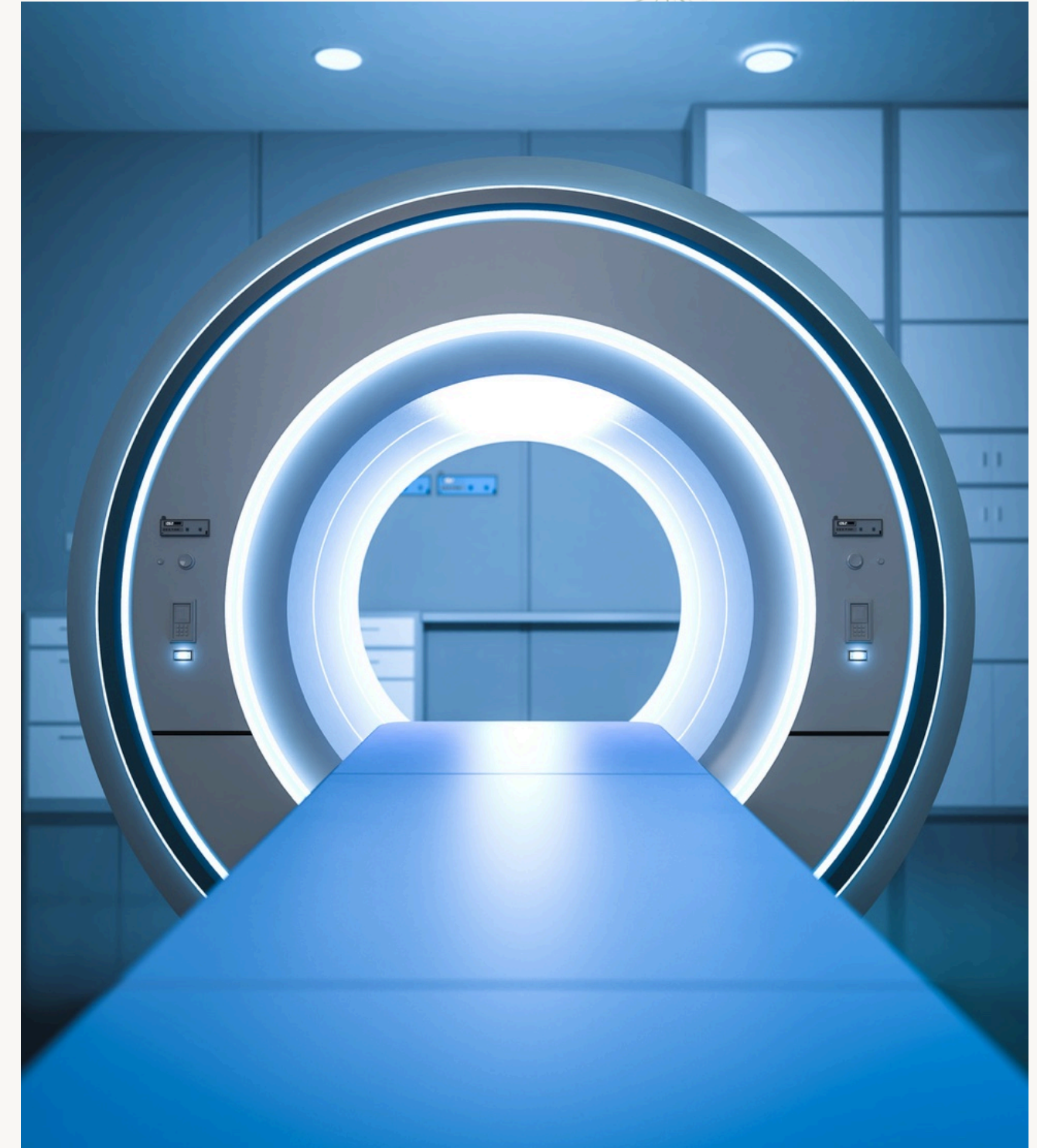


[ainhack-vandy.github.io](https://github.com/ainhack-vandy)

Energy usage of MRI Scanning

*“State-of-the-art MRI scanners require approximately **80 000–170 000 kWh** annually – equivalent to the annual energy usage of **16–34 four-person households**”*

Vosshenrich et al. (2025)



Energy usage of MRI Scanning

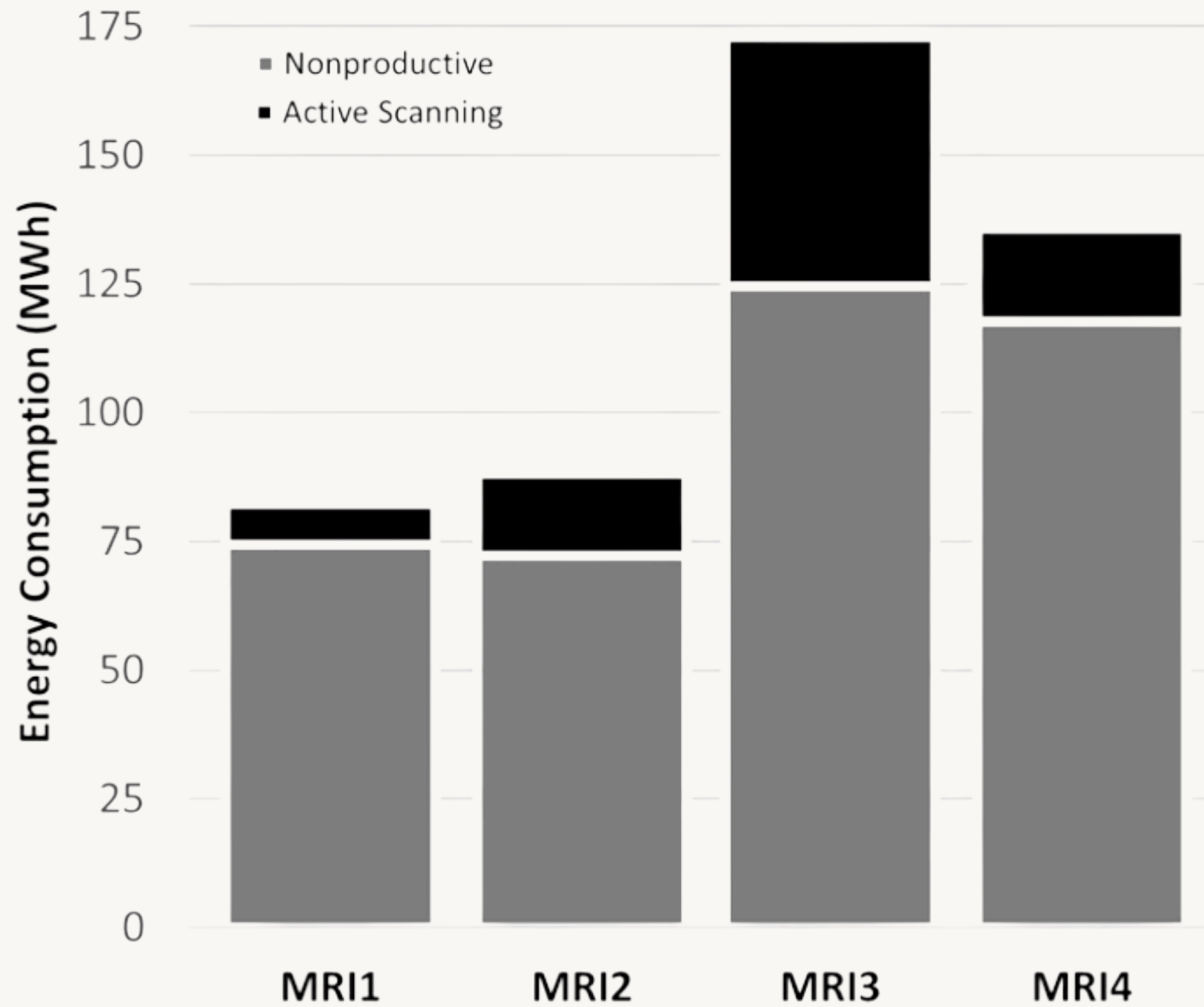


Figure 4: Projected annual energy consumption for four different MRI systems.
MWh = megawatt-hour.

72-91% is 'unproductive'

Woolen et al. (2023)

Energy usage of MRI Scanning

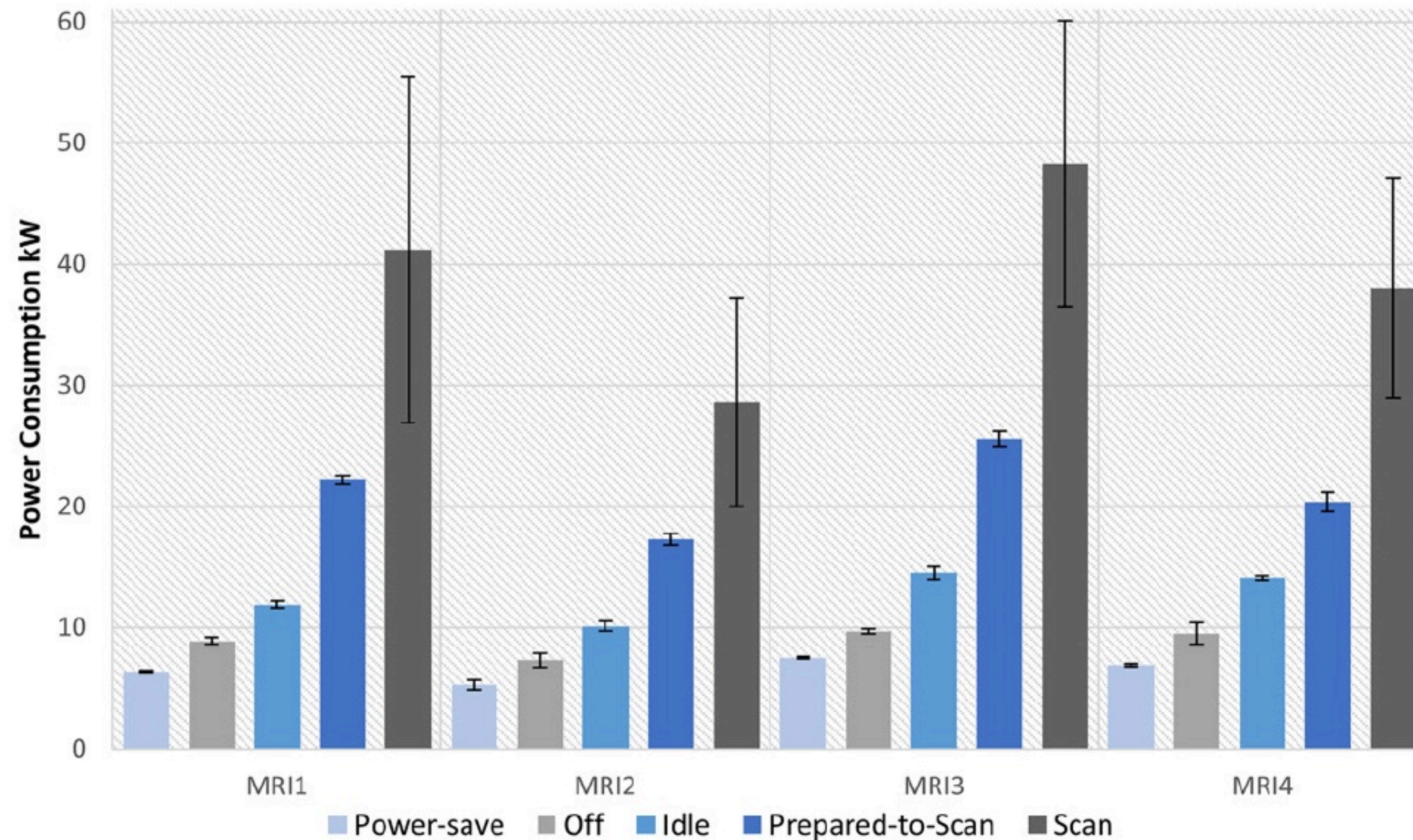


Figure 5: Power consumption of the different scanner activity modes. Power-save modes in MRI units 2–4 are simulated values based on the technique used for MRI unit 1. The error bars represent SD.

There is scope to reduce the energy usage of **MRI scanners**.

Switching scanners from ‘**idle**’ to ‘**off**’ can reduce power consumption by 25-33%.

Liquid helium for MRI scanning

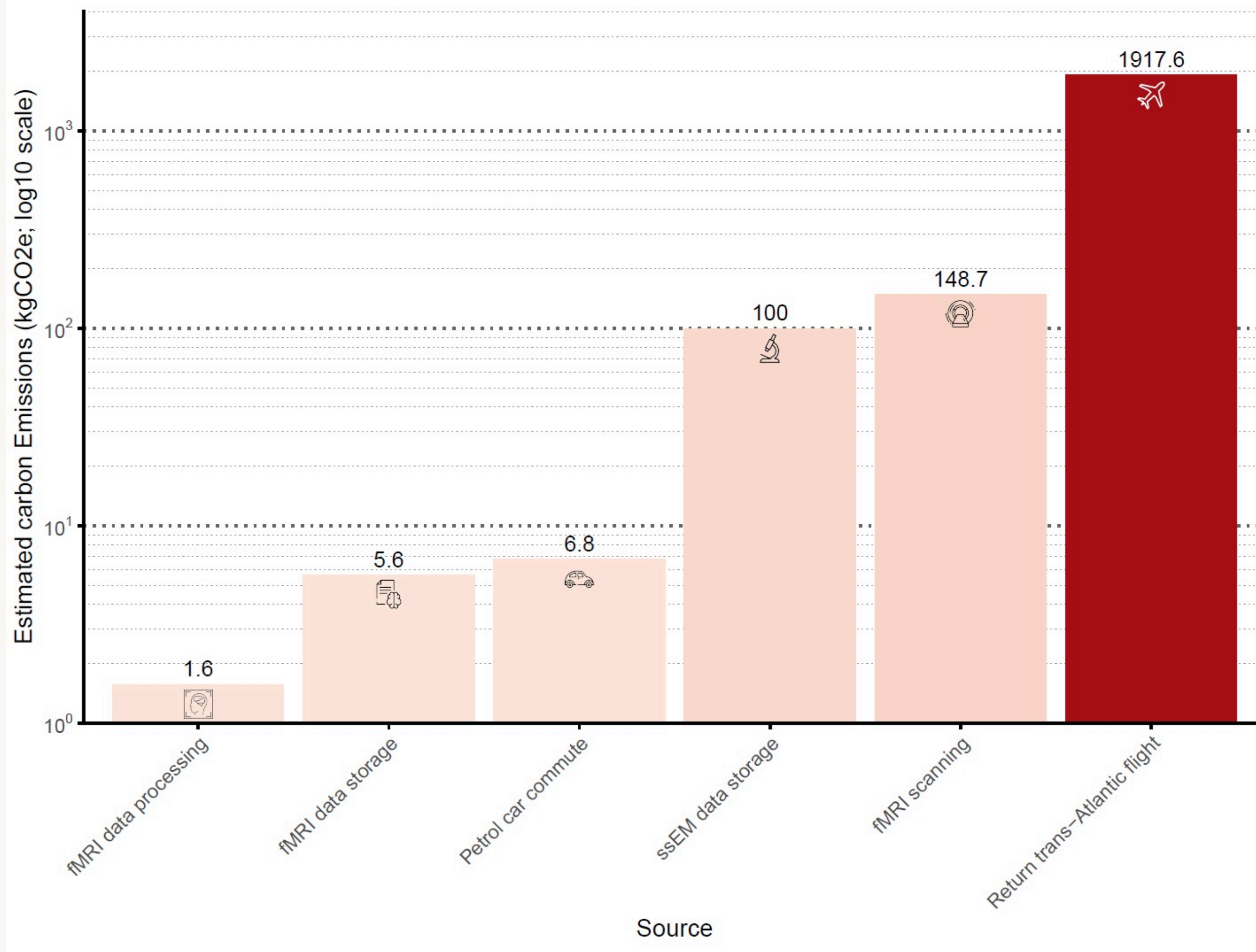


Superconductive magnets in MRI scanners require **liquid helium** for cooling. The earth's supply of liquid helium is effectively **finite and irreplaceable**.

*“**The helium shortage reached a crisis in 2012** because commercial helium production did not materialize as expected and failed to meet the helium demand worldwide, resulting in a sharp price hike.”*

Liquid helium is typically captured as a **byproduct of natural gas extraction**.

Maresh & Barker (2016)



Rae & Souter (under review)

The carbon footprint of computing

Although exact estimates vary, data centres already account for **~1-4% of global emissions**.

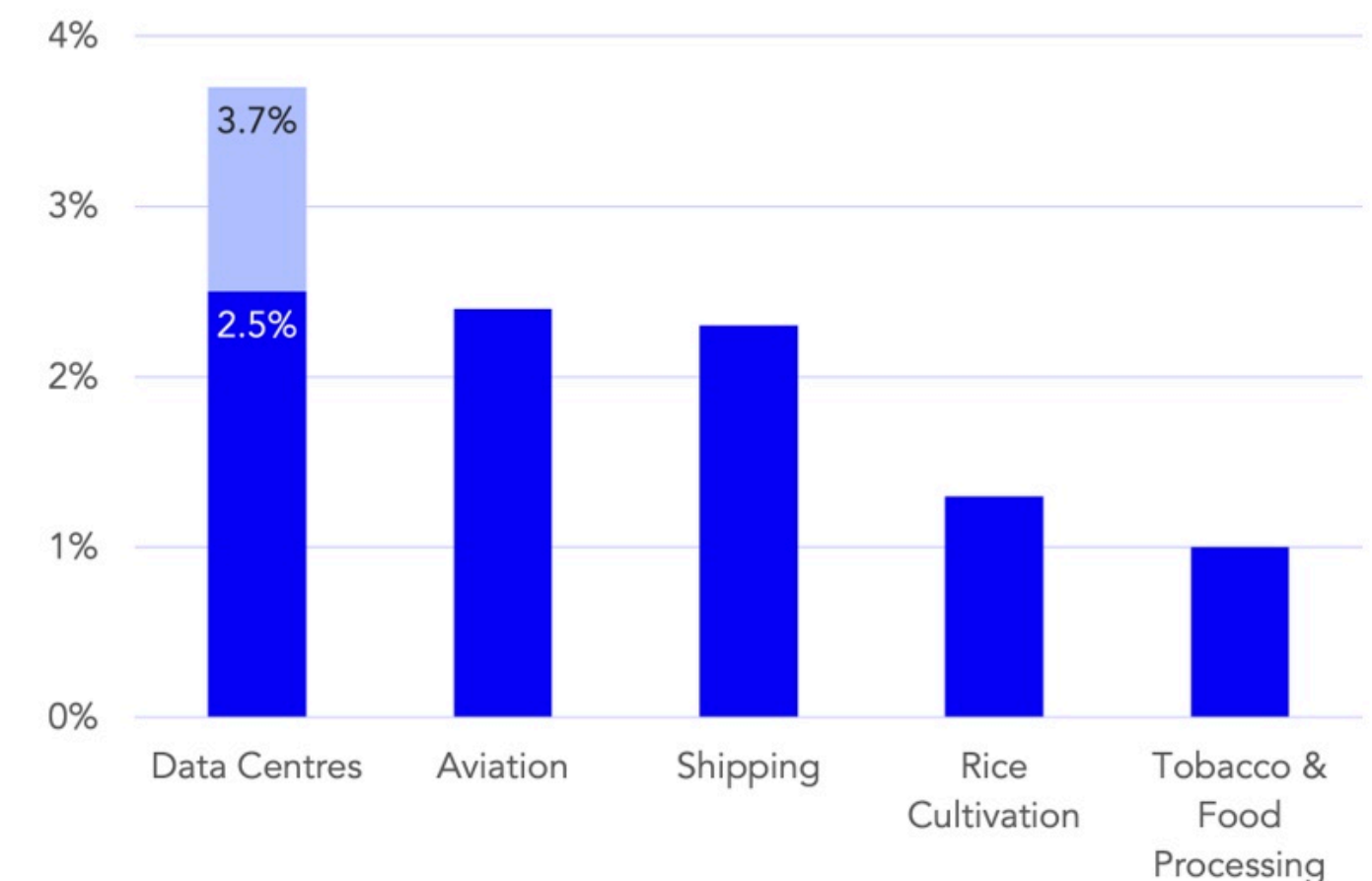
Advances in hardware mean computing is becoming more efficient.

We are like to encounter a **rebound effect** - as computing becomes cheaper, we'll use more of it.



Global cloud computing emissions exceed those from commercial aviation

Share of global CO₂ emission generated by sector/category



Source: ClimaTiq Analysis, The Shift Project, OurWorldinData



The footprint of data storage

The more **data stored** on a server, the more energy we need to operate and **cool** it.

This also provides more data to **backup**.

Importantly, the faster servers fill up, the more quickly we need to **manufacture and transport new servers**.

Dark Data & Junk Data



Dark data - Potentially usable data that is poorly indexed, and becomes functionally invisible over time.

Junk data - Data that is useless (often intermediary files) that takes up unnecessary space.

fMRIPrepCleanup

**NickESouter/
fMRIPrepCleanup**

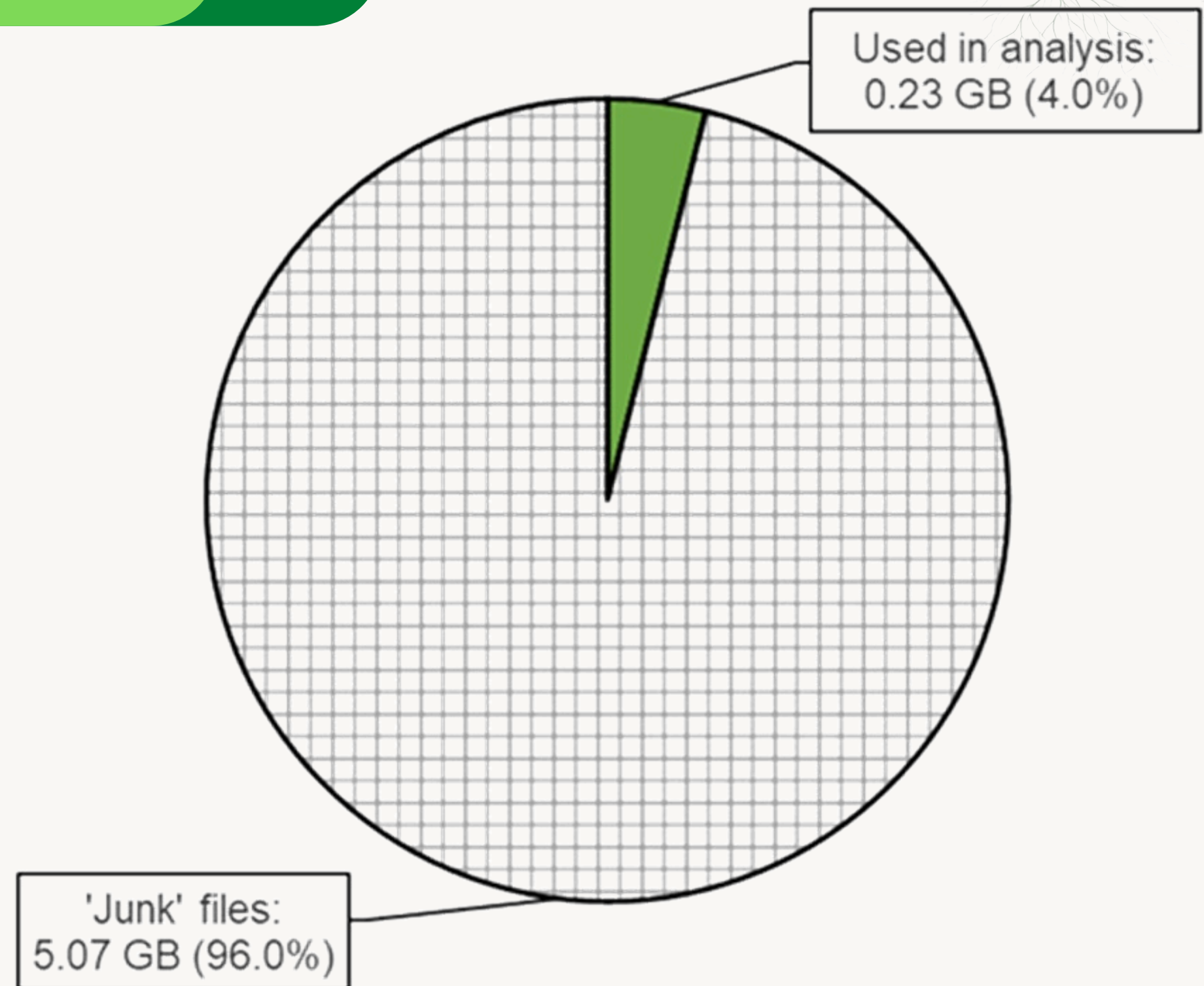
A script used to delete 'junk' files produced by fMRIPrep, to reduce storage space used by up to 95%

1 Contributor 0 Issues 2 Stars 0 Forks

NickESouter/fMRIPrepCleanup: A script used to delete 'junk' files produced by fMRIPrep, to reduce storage space used by up to 95%

A script used to delete 'junk' files produced by fMRIPrep, to reduce storage space used by up to 95% - NickESouter/fMRIPrepCleanup

GitHub



$$5.07\text{GB} \times 257 = 1.3 \text{ TB}$$

The footprint of data processing



Efficiency of Programming

Generally, **shorter runtime** corresponds to **lower energy costs**.

We can reduce our computing energy usage by **avoiding sloppy coding** (e.g., unnecessary loops within loops).

Small savings in time can add up over **many iterations**.



Table 4
Normalized global results for Energy, Time, and Memory.

Total					
	Energy (J)		Time (ms)		Mb
(c) C	1.00	(c) C	1.00	(c) Pascal	1.00
(c) Rust	1.03	(c) Rust	1.04	(c) Go	1.05
(c) C++	1.34	(c) C++	1.56	(c) C	1.17
(c) Ada	1.70	(c) Ada	1.85	(c) Fortran	1.24
(v) Java	1.98	(v) Java	1.89	(c) C++	1.34
(c) Pascal	2.14	(c) Chapel	2.14	(c) Ada	1.47
(c) Chapel	2.18	(c) Go	2.83	(c) Rust	1.54
(v) Lisp	2.27	(c) Pascal	3.02	(v) Lisp	1.92
(c) Ocaml	2.40	(c) Ocaml	3.09	(c) Haskell	2.45
(c) Fortran	2.52	(v) C#	3.14	(i) PHP	2.57
(c) Swift	2.79	(v) Lisp	3.40	(c) Swift	2.71
(c) Haskell	3.10	(c) Haskell	3.55	(i) Python	2.80
(v) C#	3.14	(c) Swift	4.20	(c) Ocaml	2.82
(c) Go	3.23	(c) Fortran	4.20	(v) C#	2.85
(i) Dart	3.83	(v) F#	6.30	(i) Hack	3.34
(v) F#	4.13	(i) JavaScript	6.52	(v) Racket	3.52
(i) JavaScript	4.45	(i) Dart	6.67	(i) Ruby	3.97
(v) Racket	7.91	(v) Racket	11.27	(c) Chapel	4.00
(i) TypeScript	21.50	(i) Hack	26.99	(v) F#	4.25
(i) Hack	24.02	(i) PHP	27.64	(i) JavaScript	4.59
(i) PHP	29.30	(v) Erlang	36.71	(i) TypeScript	4.69
(v) Erlang	42.23	(i) Jruby	43.44	(v) Java	6.01
(i) Lua	45.98	(i) TypeScript	46.20	(i) Perl	6.62
(i) Jruby	46.54	(i) Ruby	59.34	(i) Lua	6.72
(i) Ruby	69.91	(i) Perl	65.79	(v) Erlang	7.20
(i) Python	75.88	(i) Python	71.90	(i) Dart	8.64
(i) Perl	79.58	(i) Lua	82.91	(i) Jruby	19.84

Efficiency of Programming



CPU Energy Usage

$$\frac{\text{CPU time} \times \text{thermal design power}}{\text{number of CPUs}}$$

RAM Energy Usage

Maximum memory used
x
volatile memory consumption
x
runtime

Power Use Effectiveness (PUE)

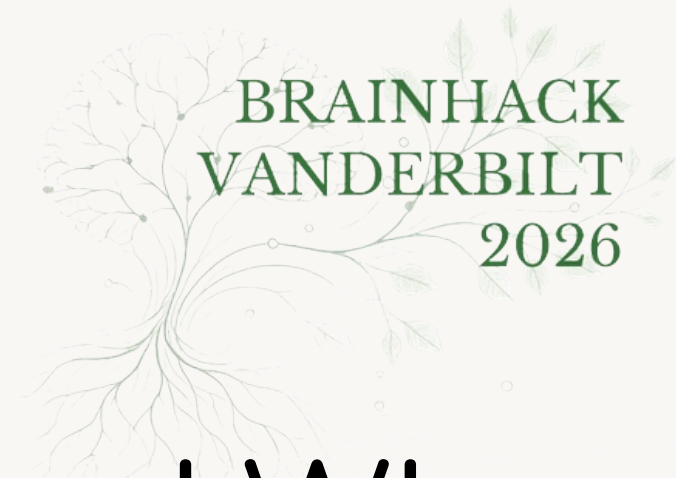
Reflects **energy overheads** (e.g., lighting, cooling) used in an HPC data centre.

Sussex HPC currently has **PUE of 1.28**, better than industry standard (**1.55**).

For each **1 kWh** used for computing, **0.28 kWh** are used for overheads.



Carbon Intensity



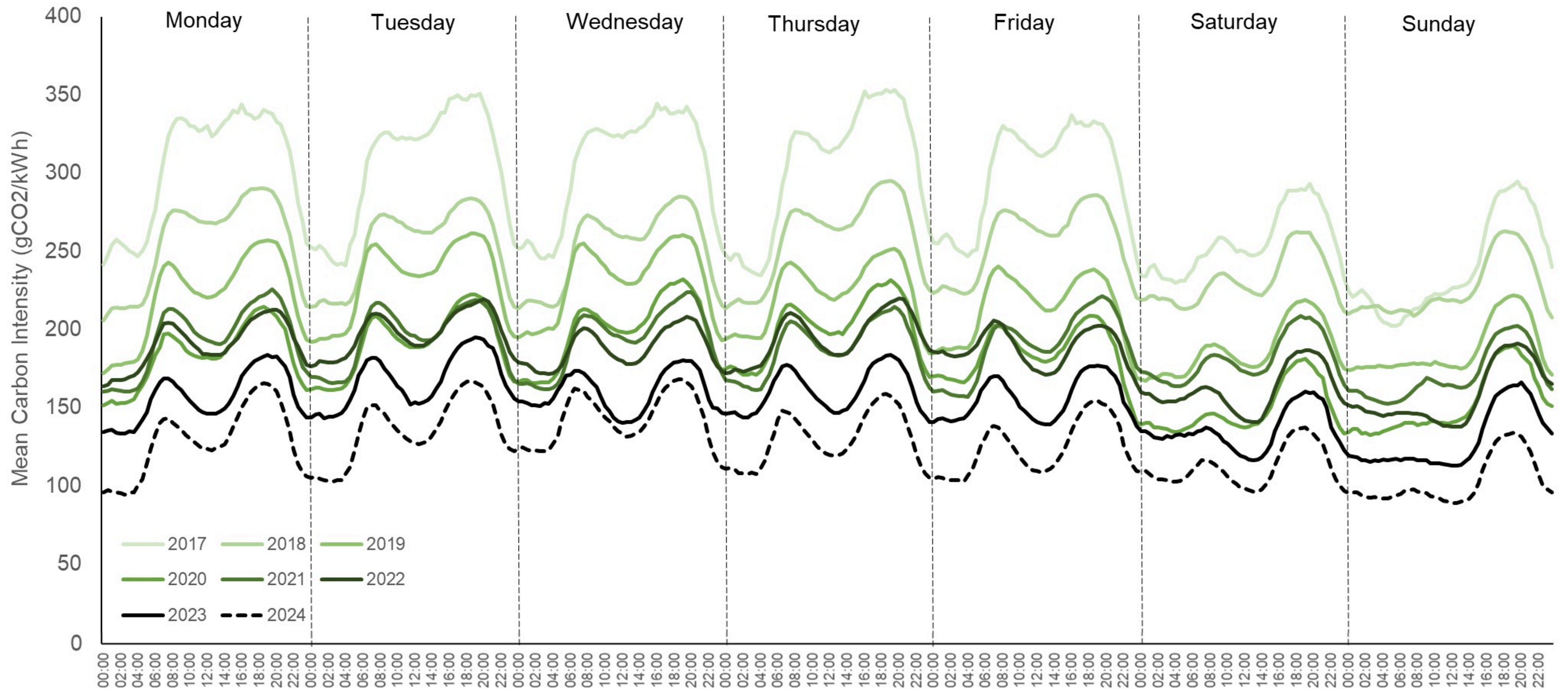
Grams of carbon dioxide equivalent emissions per kWh of energy (gCO₂/kWh). Lower when we use more renewable energy sources.

Live data for the UK: <https://carbonintensity.org.uk/>

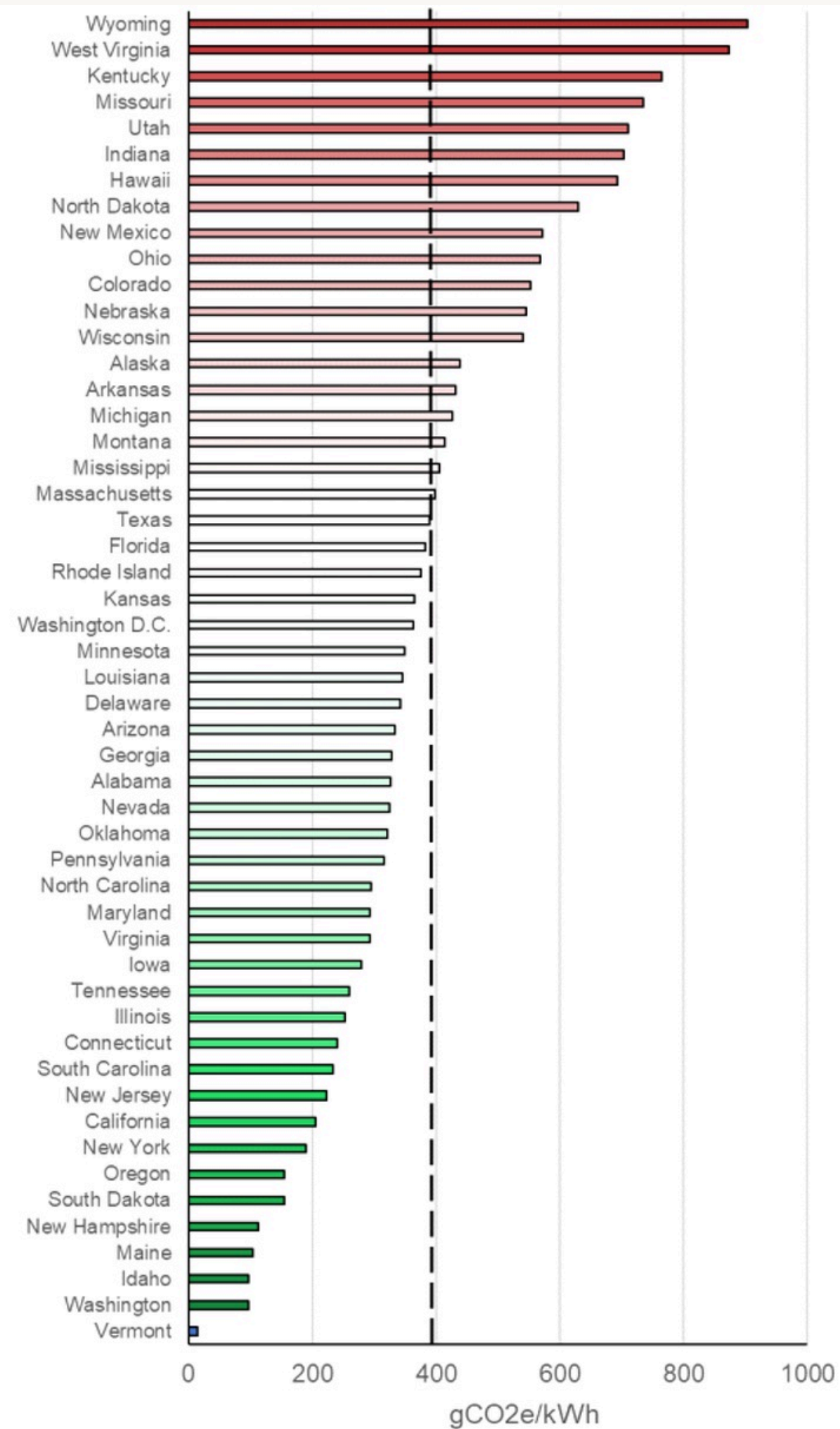
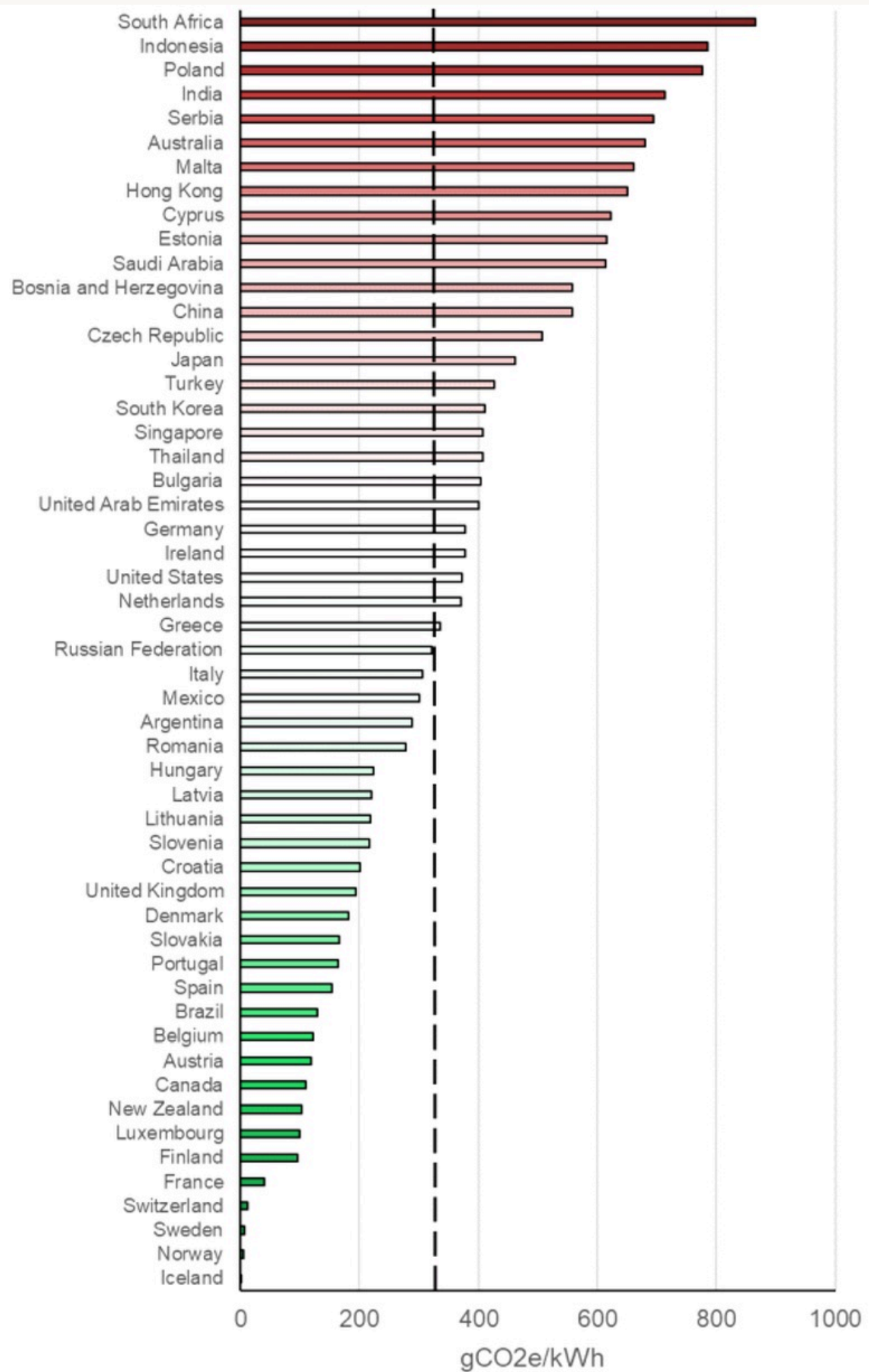


Carbon Intensity

Souter et al. (2023),
Imaging Neuroscience [Updated]



Carbon Intensity



Souter et al. (2023),
Imaging Neuroscience

Carbon Emissions

$$\begin{aligned} &(\text{CPU} + \text{RAM energy}) \\ &\times \\ &\text{Power use effectiveness} \\ &\times \\ &\text{Carbon intensity} \end{aligned}$$

Carbon Tracking

Table 1 | Pros and cons of each strategy for carbon footprint estimation

Characteristic of each strategy	Online calculator	Embedded package	Server-side tool
Compatible with any hardware	Yes	No, a dedicated tool needs to be built for each	Yes, provided hardware usage can be monitored
Compatible with any programming language	Yes		Yes
Compatible with any task or research field	Yes		Yes
Computing metrics are collected automatically	No	Yes	Yes
Does not interfere with existing code	Yes	No	Yes
Estimates can be obtained beforehand	Yes	No	No
Estimates can be obtained in real time	No	Yes	Yes
Estimates can be obtained retrospectively	Yes	Only if the tracker was active at the time	Yes
Scalable with large numbers of jobs	No	Yes	Yes
Scalable over long periods of time	No	Only if the tracker is used every time	Yes

Lannelongue & Inouye (2023),
Nature Reviews Methods Primers



Carbon Tracking Green Algorithms

Green Algorithms
How green are your computations?

Check out the new **Green Algorithms website**: www.green-algorithms.org.

Details about your algorithm
To understand how each parameter impacts your carbon footprint, check out the formula below and the [methods article](#)

Runtime (HH:MM)

Type of cores

CPUS

Number of cores

Model

CO₂ ... g CO₂e
Carbon footprint

Energy needed

Carbon sequestration

0.00 km
in a passenger car

0
flights None

<https://calculator.green-algorithms.org/>

Carbon Tracking

Green Algorithms



14.91 g CO₂e
Carbon footprint



134.14 Wh
Energy needed



1.63e-02 tree-
months
Carbon sequestration



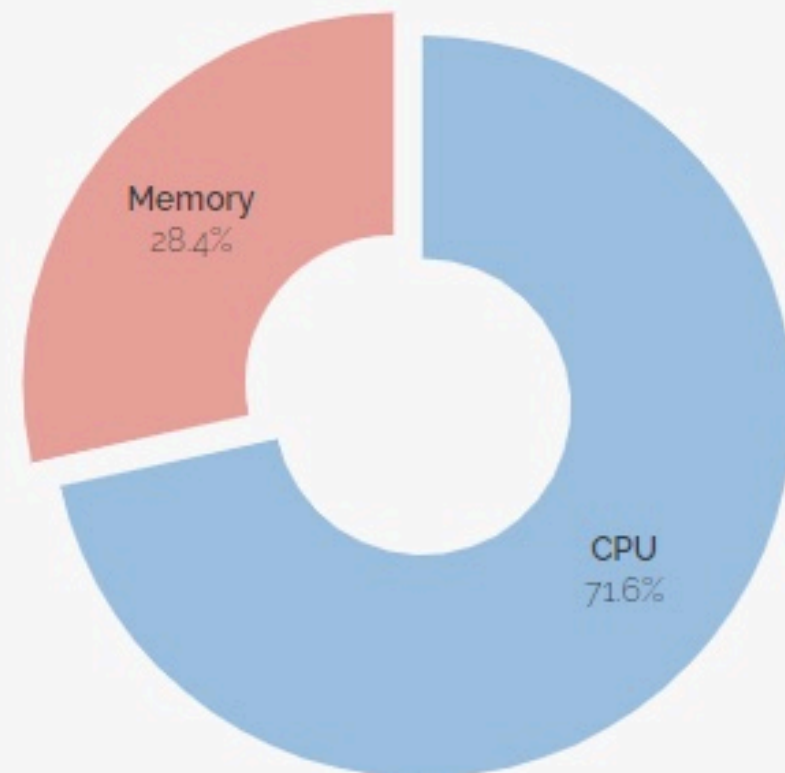
8.52e-02 km
in a passenger car



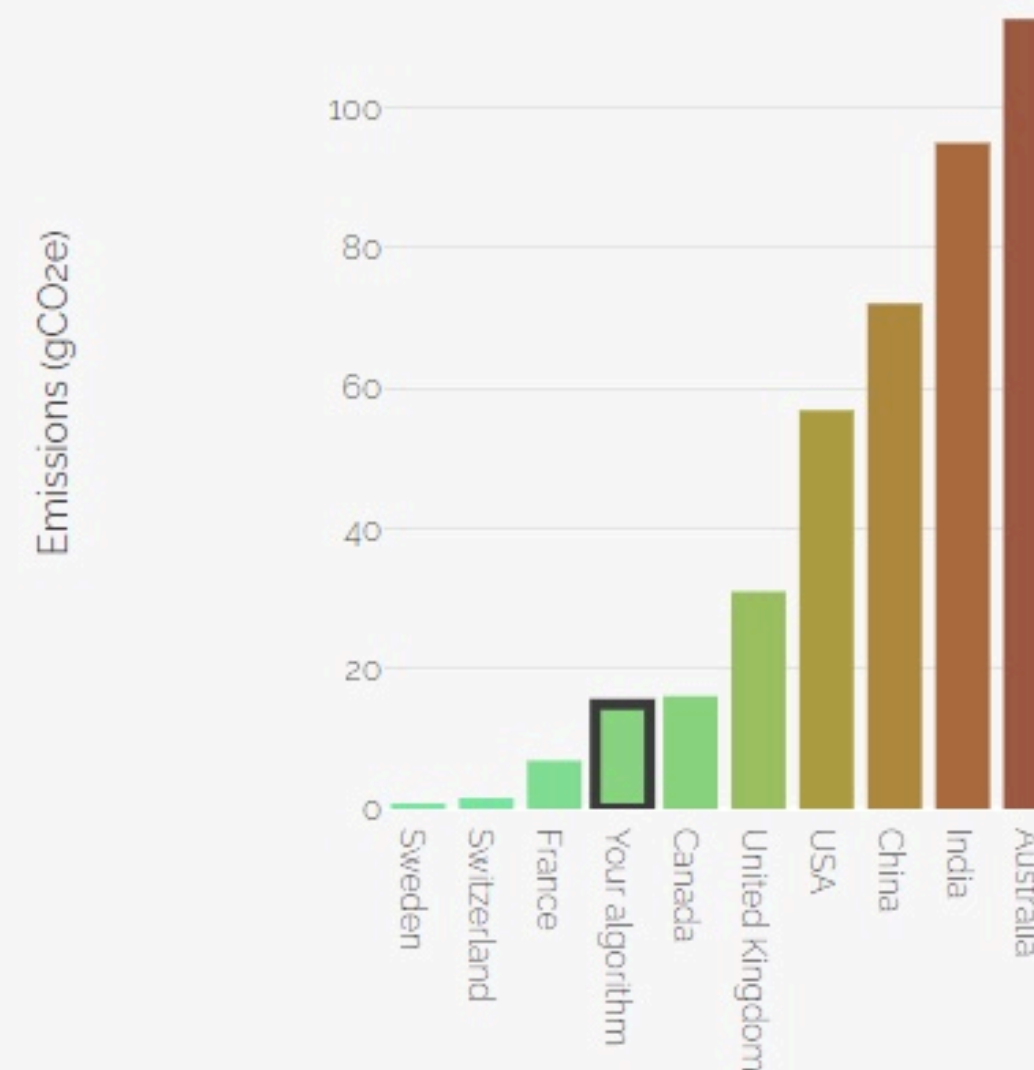
0.03%
of a flight Paris-London

Carbon Tracking Green Algorithms

Computing cores VS Memory



How the location impacts your footprint





Reducing Use: Case studies from neuroimaging

**Measuring and reducing
the carbon footprint of
fMRI preprocessing
in fMRIPrep**

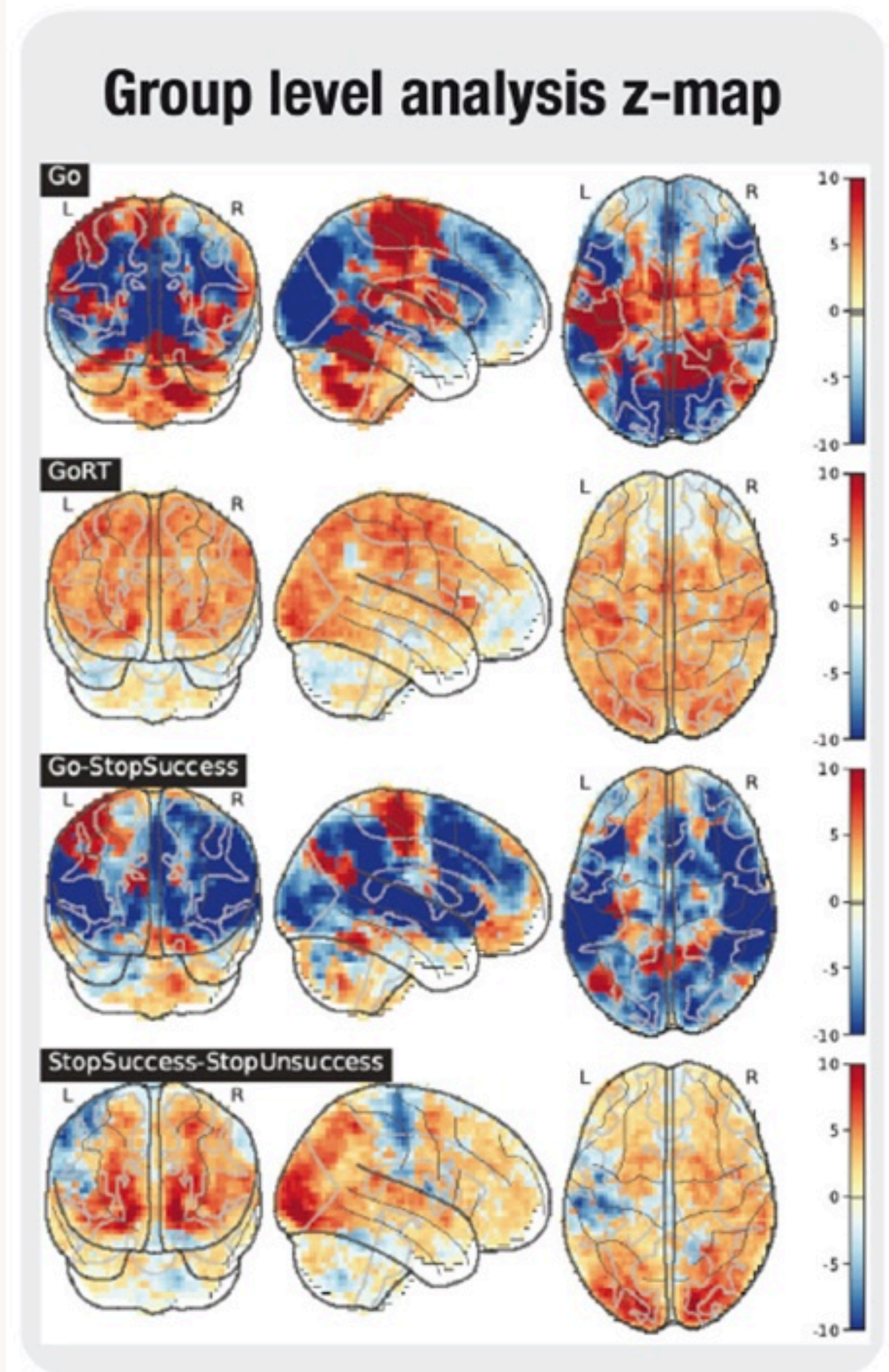

```
usage: fmriprep [-h] [--version] [--skip_bids_validation]
               [--participant-label PARTICIPANT_LABEL [PARTICIPANT_LABEL ...]]
               [-t TASK_ID] [--echo-idx ECHO_IDX] [--bids-filter-file FILE]
               [--anat-derivatives PATH] [--bids-database-dir PATH]
               [--nprocs NPROCS] [--omp-nthreads OMP_NTHREADS]
               [--mem MEMORY_MB] [--low-mem] [--use-plugin FILE]
               [--anat-only] [--boilerplate_only] [--md-only-boilerplate]
               [--error-on-aroma-warnings] [-v]
               [--ignore {fieldmaps,slicetiming,sbref,t2w,flair}
               [{fieldmaps,slicetiming,sbref,t2w,flair} ...]]
               [--longitudinal] [--output-spaces [OUTPUT_SPACES ...]]
               [--me-output-echos] [--bold2t1w-init {register,header}]
               [--bold2t1w-dof {6,9,12}] [--force-bbr] [--force-no-bbr]
               [--medial-surface-nan] [--slice-time-ref SLICE_TIME_REF]
               [--dummy-scans DUMMY_SCANS] [--random-seed _RANDOM_SEED]
               [--use-aroma]
               [--aroma-melodic-dimensionality AROMA_MELODIC_DIM]
               [--return-all-components]
               [--fd-spike-threshold REGRESSORS_FD_TH]
               [--dvars-spike-threshold REGRESSORS_DVARS_TH]
               [--skull-strip-template SKULL_STRIP_TEMPLATE]
               [--skull-strip-fixed-seed]
               [--skull-strip-t1w {auto,skip,force}] [--fmap-bspline]
               [--fmap-no-demean] [--topup-max-vols TOPUP_MAX_VOLS]
               [--use-syn-sdc [{warn,error}]] [--force-syn]
               [--fs-license-file FILE] [--fs-subjects-dir PATH]
               [--no-submm-recon] [--cifti-output [{91k,170k}] |
               --fs-no-reconall] [--output-layout {bids,legacy}]
               [-w WORK_DIR] [--clean-workdir] [--resource-monitor]
               [--reports-only] [--config-file FILE] [--write-graph]
```

fmriprep

The Data

UCLA Consortium for
Neuropsychiatric Phenomics
LA5c Study

1 run of a stop signal task
for 257 subjects



Measuring Performance

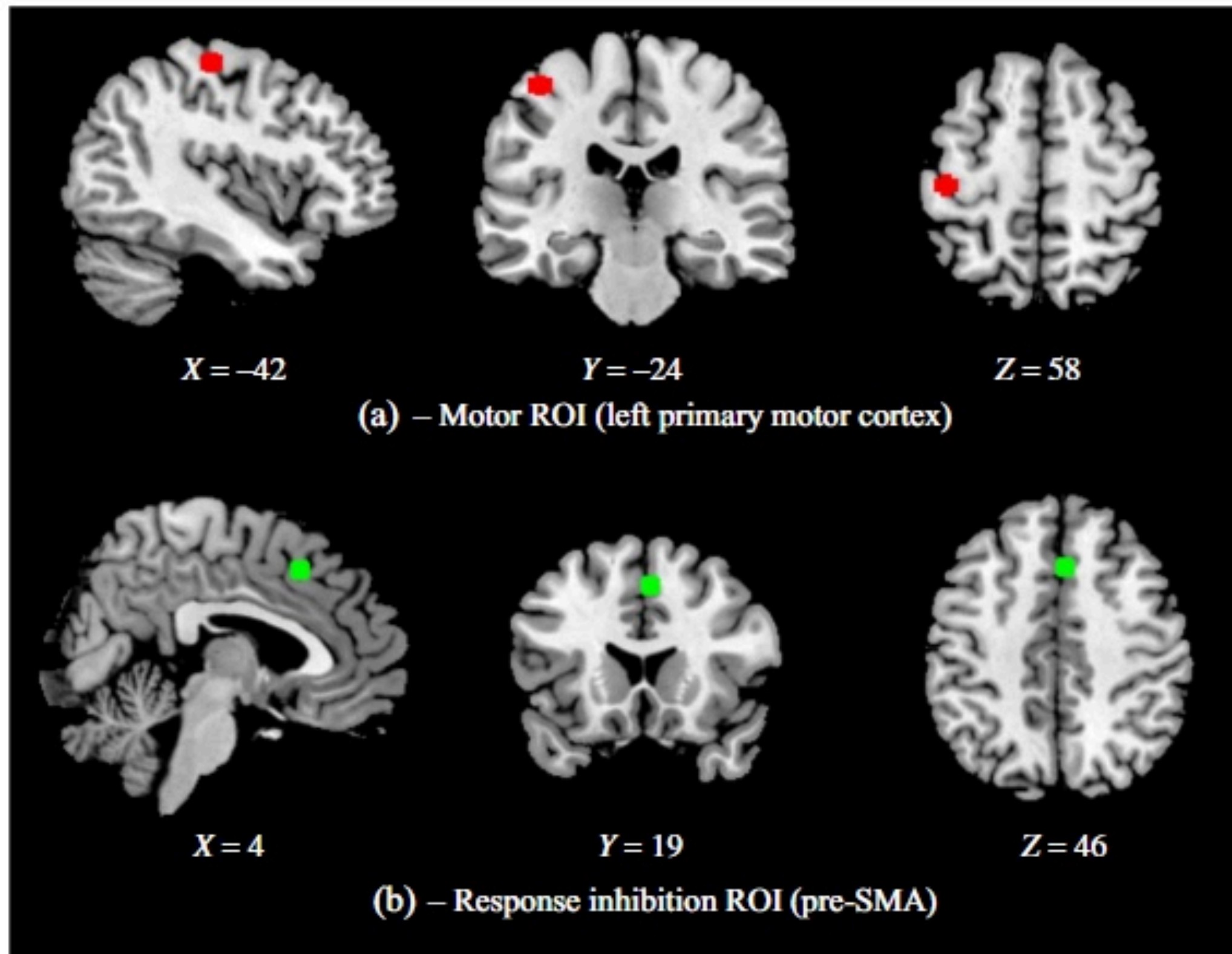


FIGURE 1 ROIs for use in FSL Featquery, including the left primary motor cortex for motor response activation, and the pre-SMA for successful response inhibition activation. pre-SMA, pre-supplementary motor area; ROI, region of interest.

fMRIPrep pipeline variants



P0: Baseline

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

**P2: 'Sloppy' testing
mode**

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

**P2: 'Sloppy' testing
mode**

**P3: Low memory
mode**

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

**P2: 'Sloppy' testing
mode**

**P3: Low memory
mode**

**P4: Add surface
output space**

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

**P2: 'Sloppy' testing
mode**

**P3: Low memory
mode**

**P4: Add surface
output space**

P5: ICA-AROMA

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

**P2: 'Sloppy' testing
mode**

**P3: Low memory
mode**

**P4: Add surface
output space**

P5: ICA-AROMA

**P6: Fieldmap-free
distortion correction**

fMRIPrep pipeline variants



P0: Baseline

**P1: No FreeSurfer
surface reconstruction**

**P2: 'Sloppy' testing
mode**

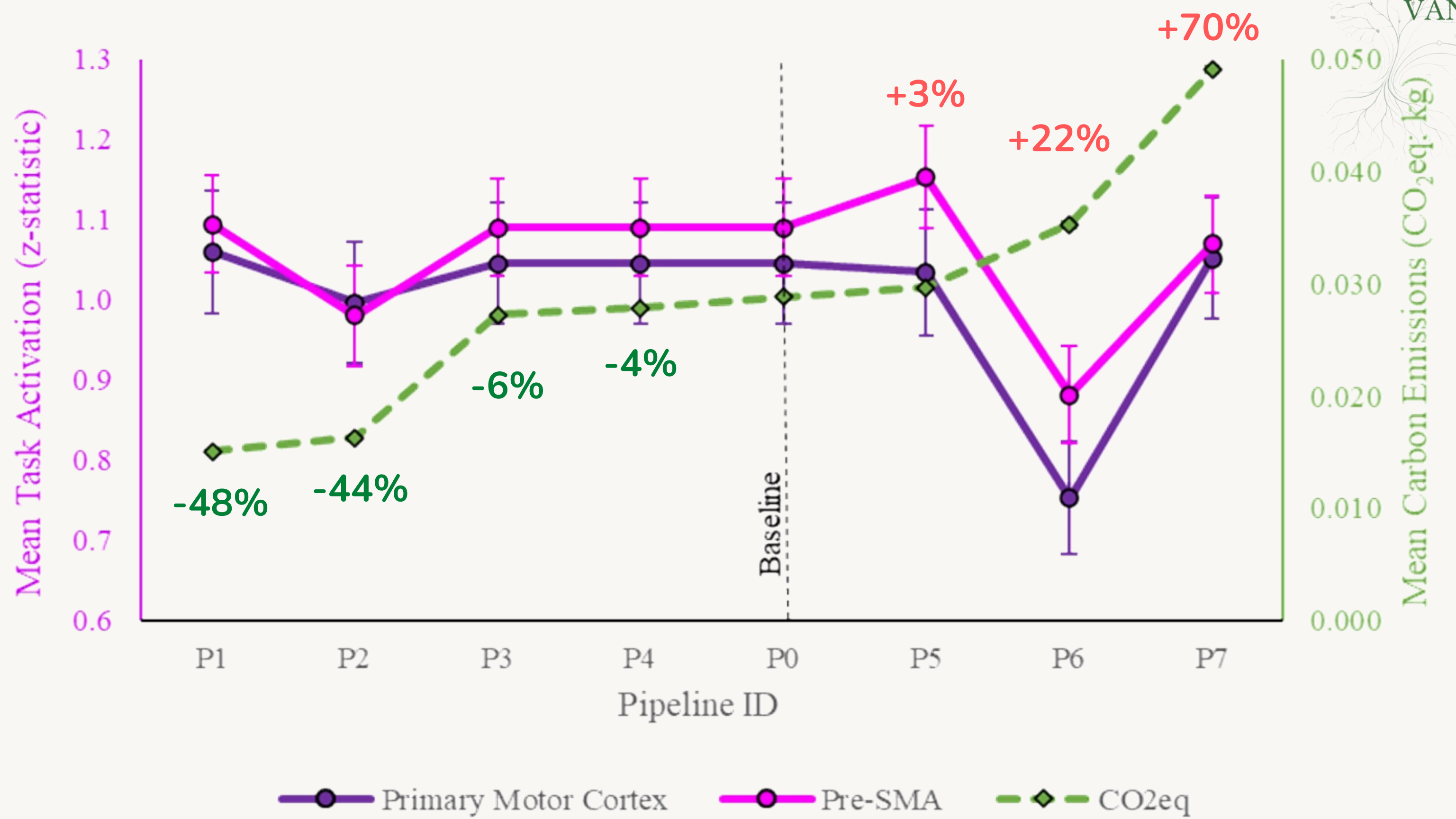
**P3: Low memory
mode**

**P4: Add surface
output space**

P5: ICA-AROMA

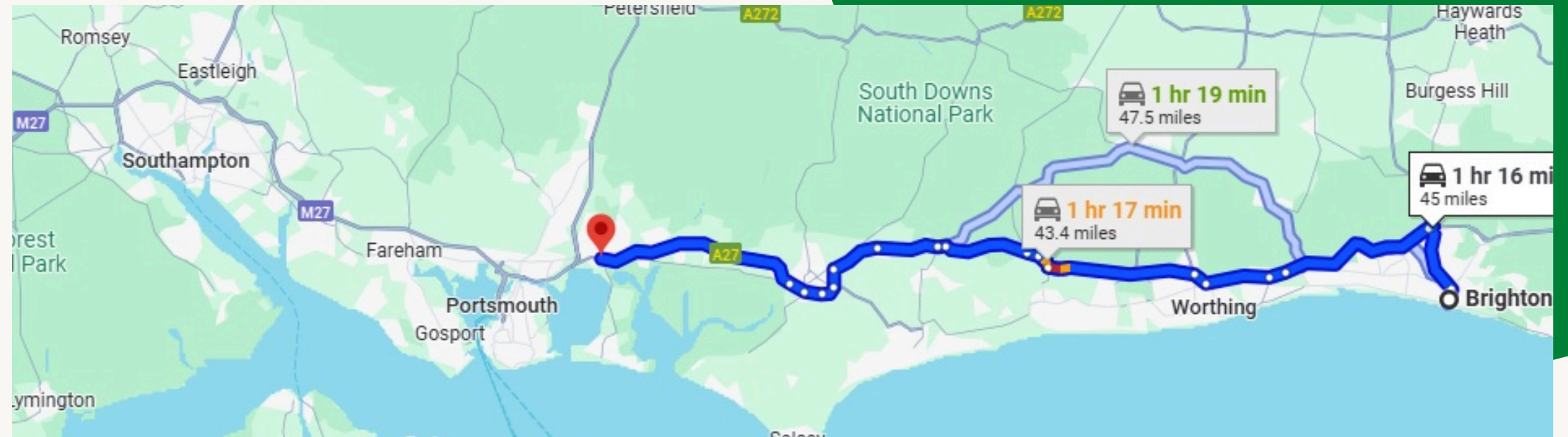
**P6: Fieldmap-free
distortion correction**

**P7: Increased output
space resolution**



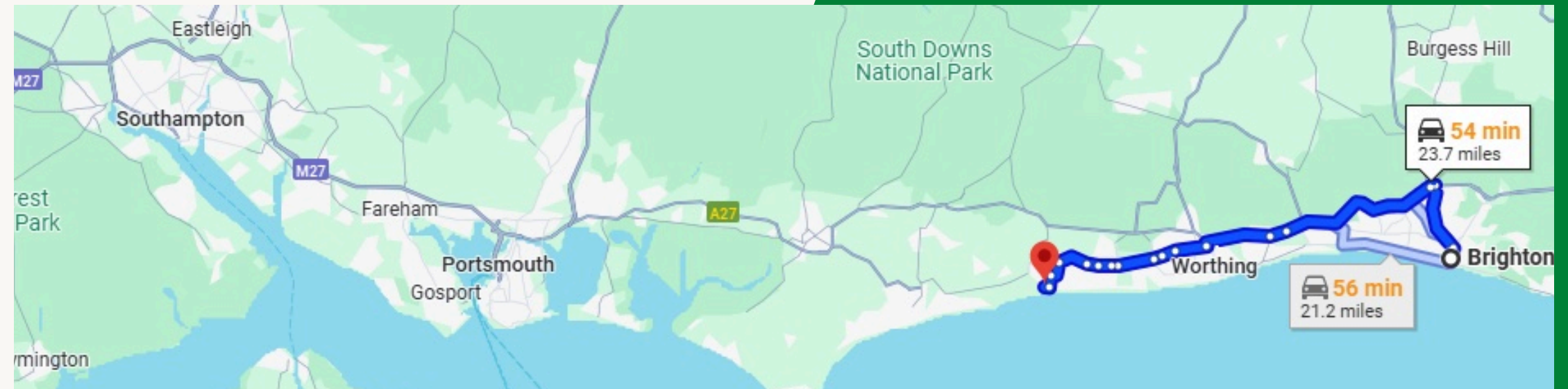
Road trip

P0 - Baseline
7.5 kg = 43 miles
Portsmouth

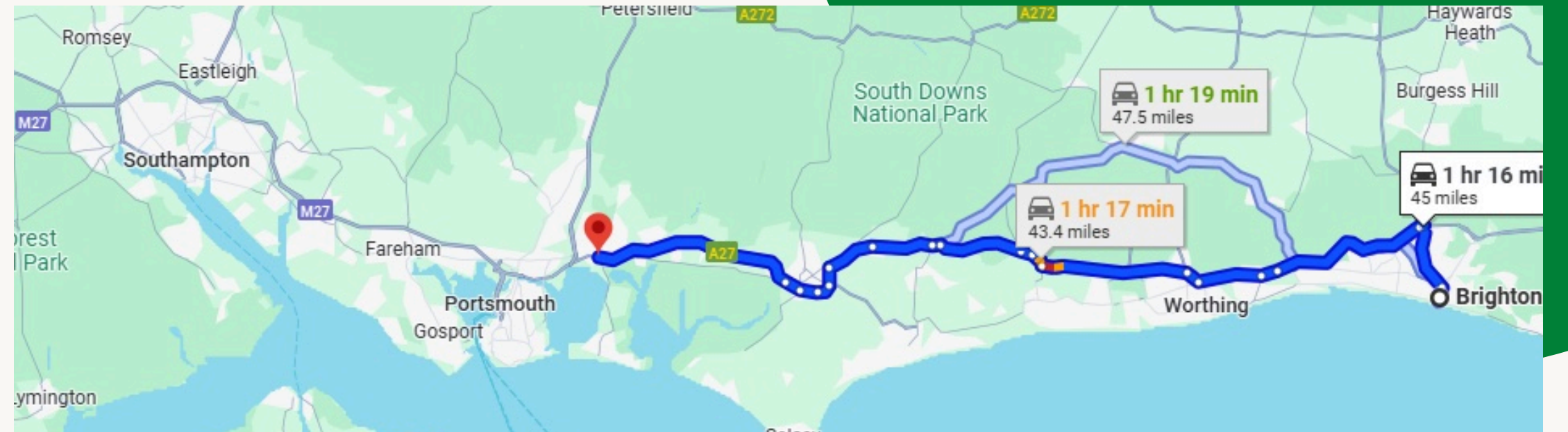


Road trip

P1 - No FreeSurfer
3.9 kg = 22 miles
Littlehampton

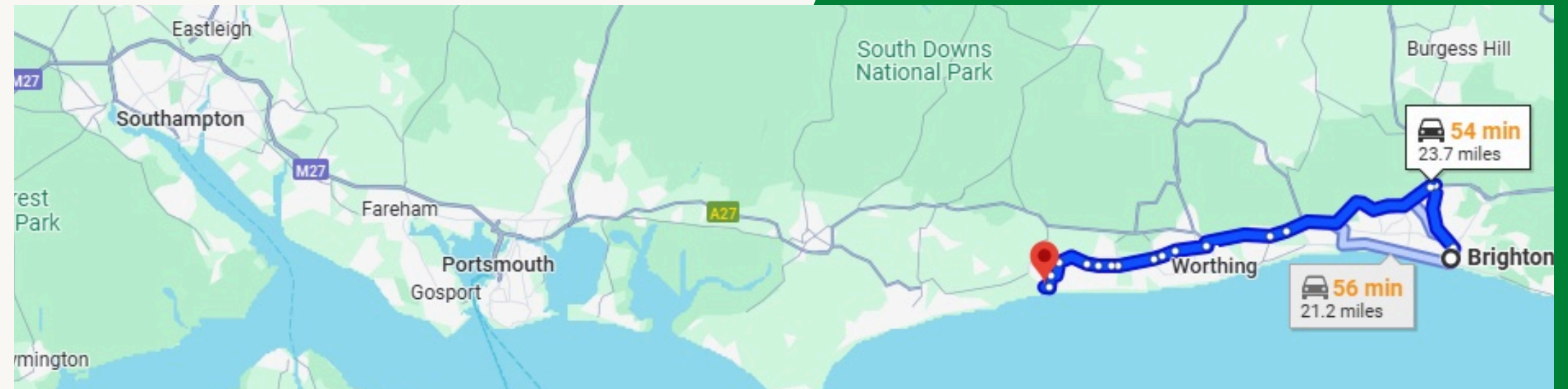


P0 - Baseline
7.5 kg = 43 miles
Portsmouth

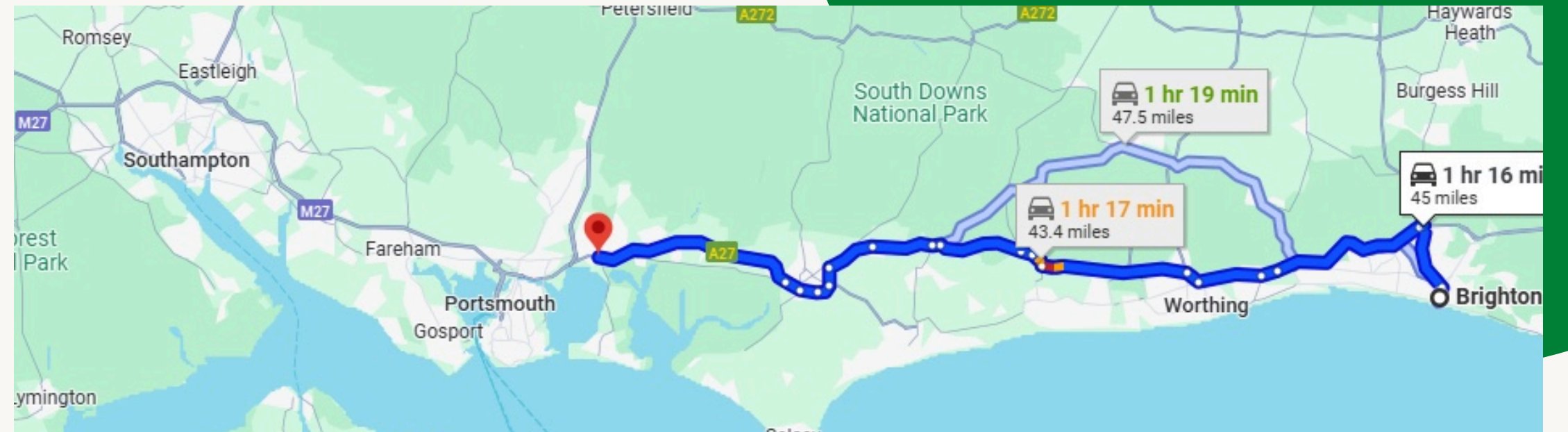


Road trip

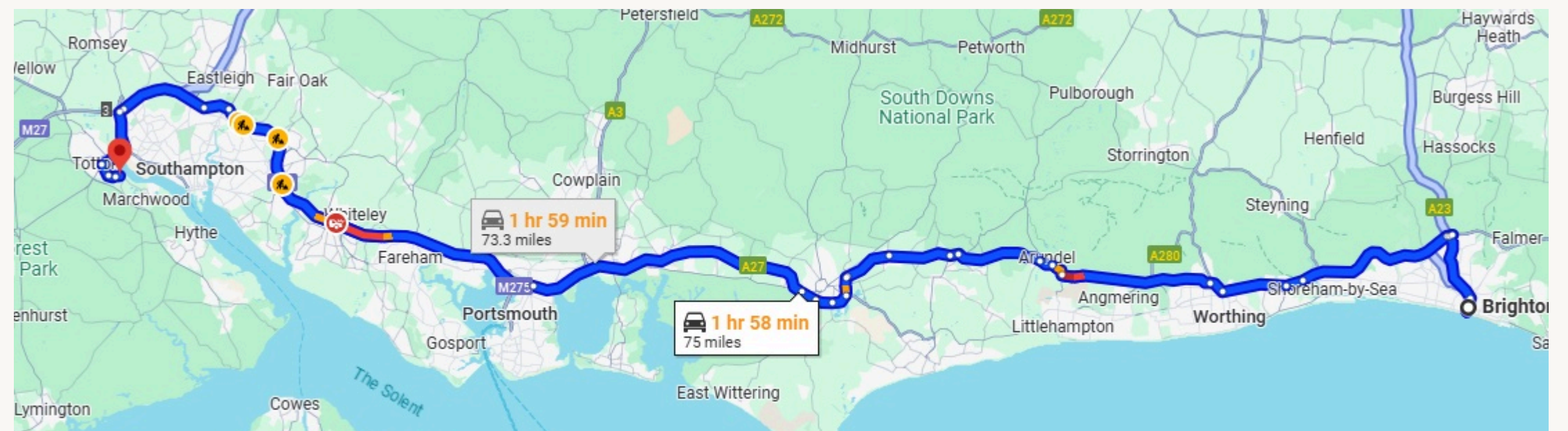
P1 - No FreeSurfer
3.9 kg = 22 miles
Littlehampton



P0 - Baseline
7.5 kg = 43 miles
Portsmouth



P7 - High Resolution
12.7 kg = 73 miles
Southampton



Recommendations for fMRIPrep



1

Disable FreeSurfer surface reconstruction (`--fs-no-reconall`) if reconstruction files are not required.

Recommendations for fMRIPrep

1

Disable FreeSurfer surface reconstruction (`--fs-no-reconall`) if reconstruction files are not required.

2

Only when testing the feasibility of a pipeline, use 'sloppy' registration (`--sloppy`).

Recommendations for fMRIPrep

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Disable FreeSurfer surface reconstruction (`--fs-no-reconall`) if reconstruction files are not required.

2

Only when testing the feasibility of a pipeline, use 'sloppy' registration (`--sloppy`).

3

Implement low memory mode (`--low-mem`).

Recommendations for fMRIPrep

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Disable FreeSurfer surface reconstruction (`--fs-no-reconall`) if reconstruction files are not required.

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Only when testing the feasibility of a pipeline, use 'sloppy' registration (`--sloppy`).

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Implement ICA-AROMA (`--use-aroma`) if needed.

Recommendations for fMRIPrep



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Disable FreeSurfer surface reconstruction (`--fs-no-reconall`) if reconstruction files are not required.

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Only when testing the feasibility of a pipeline, use 'sloppy' registration (`--sloppy`).

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Implement low memory mode (`--low-mem`).

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Implement ICA-AROMA (`--use-aroma`) if needed.

5

Be aware that fieldmap-free distortion correction technique (`--use-syn-sdc`) can both increase emissions and degrade data quality.

Recommendations for fMRIPrep



1

Disable FreeSurfer surface reconstruction (`--fs-no-reconall`) if reconstruction files are not required.

2

Only when testing the feasibility of a pipeline, use 'sloppy' registration (`--sloppy`).

3

Implement low memory mode (`--low-mem`).

4

Implement ICA-AROMA (`--use-aroma`) if needed.

5

Be aware that fieldmap-free distortion correction technique (`--use-syn-sdc`) can both increase emissions and degrade data quality.

6

Refrain from using high-resolution volumetric output space templates (`--output-spaces MNI152NLin6Asym:res-1`) unless necessary.

The background features a dark blue field with a network of white nodes connected by blue lines, resembling a data graph. On the left side, there is a cluster of white gears of various sizes. A green diagonal band runs from the top left towards the bottom right, separating the title area from the subtitle area.

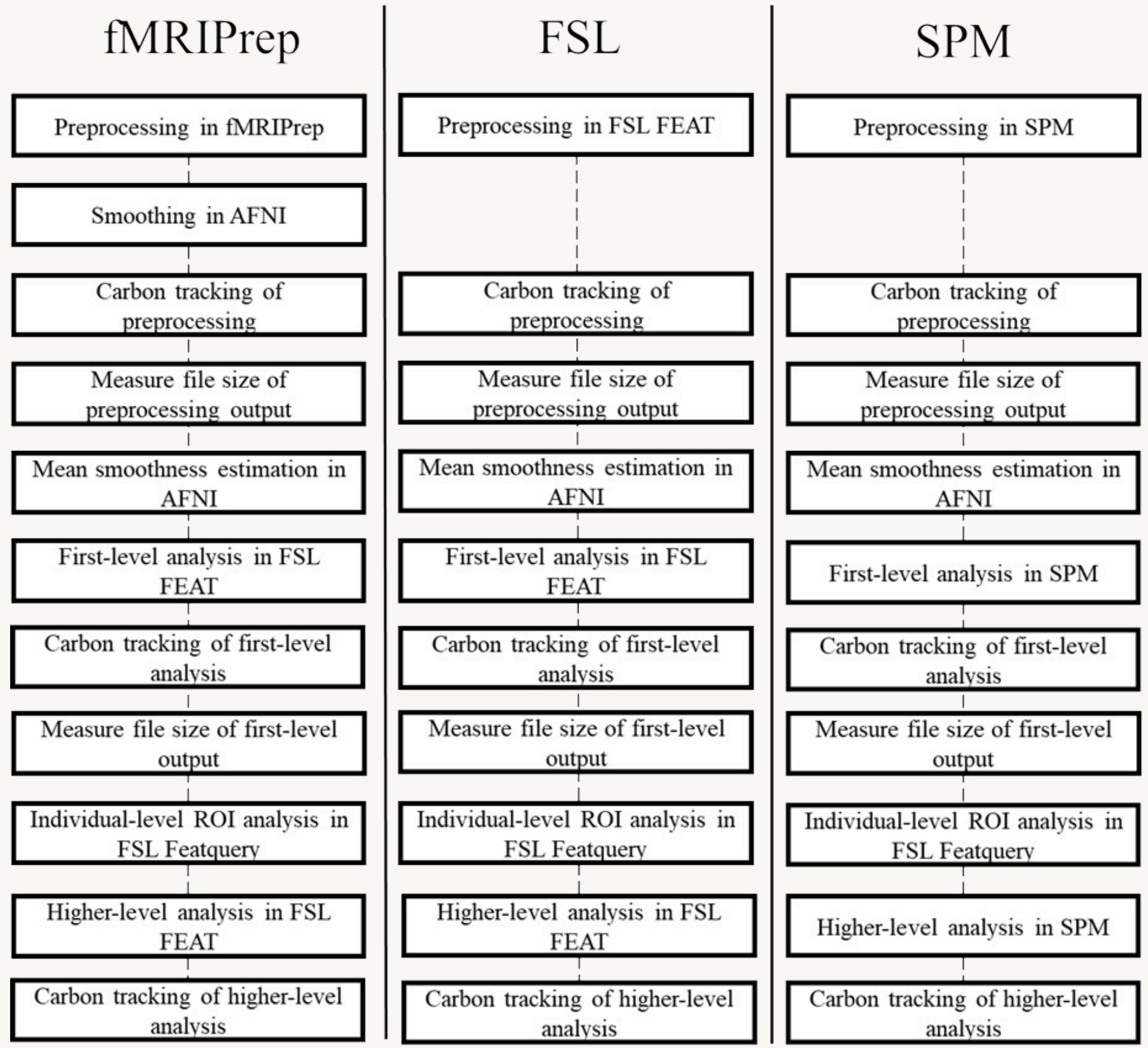
Reducing Use: Case studies from neuroimaging

Comparing the carbon footprint of fMRI data processing and analysis approaches

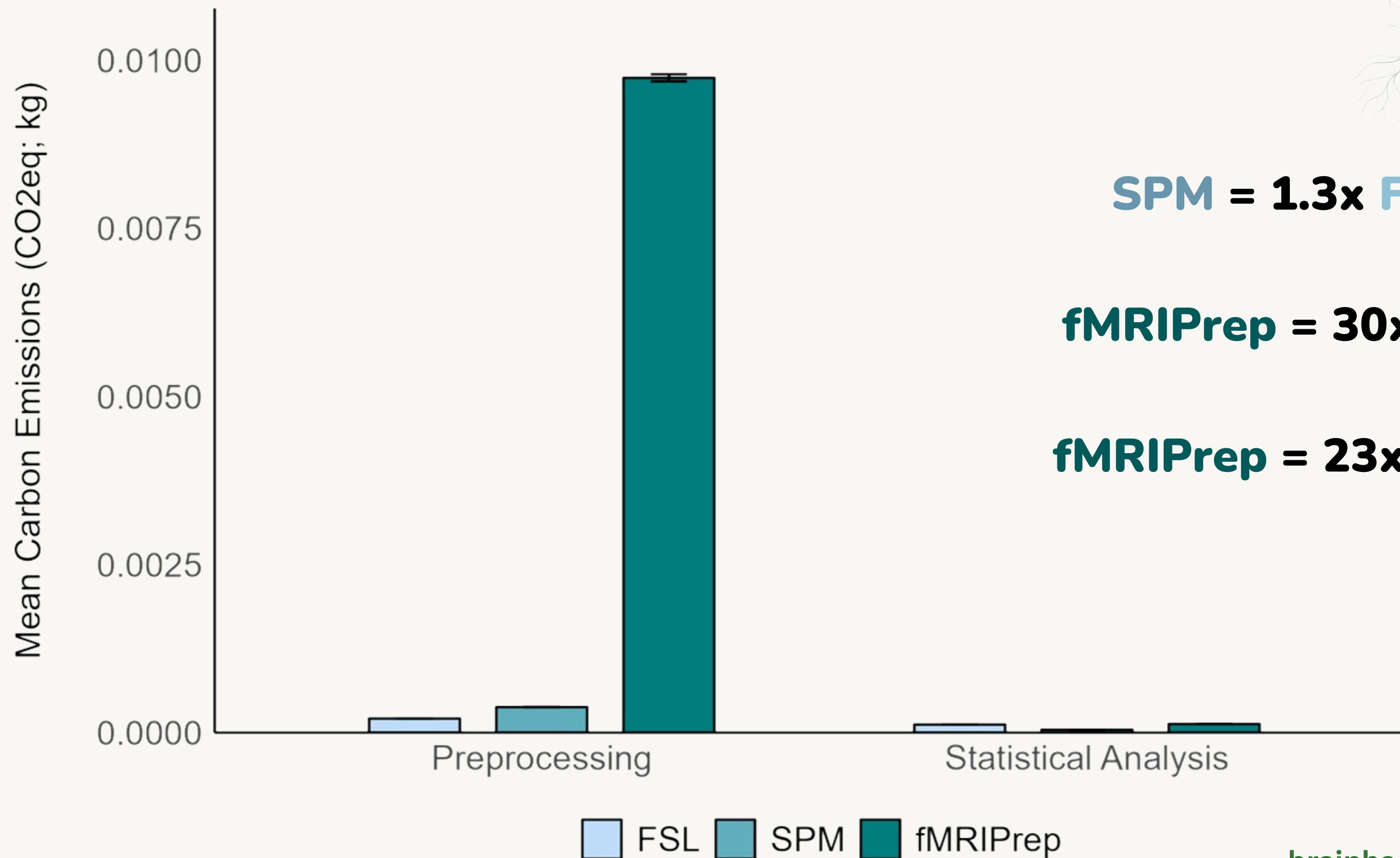
Comparing Packages

How does the footprint of computing vary across **FSL**, **SPM**, and **fMRIPrep**?

Do higher emissions correspond with higher quality output?



Carbon Emissions by Package

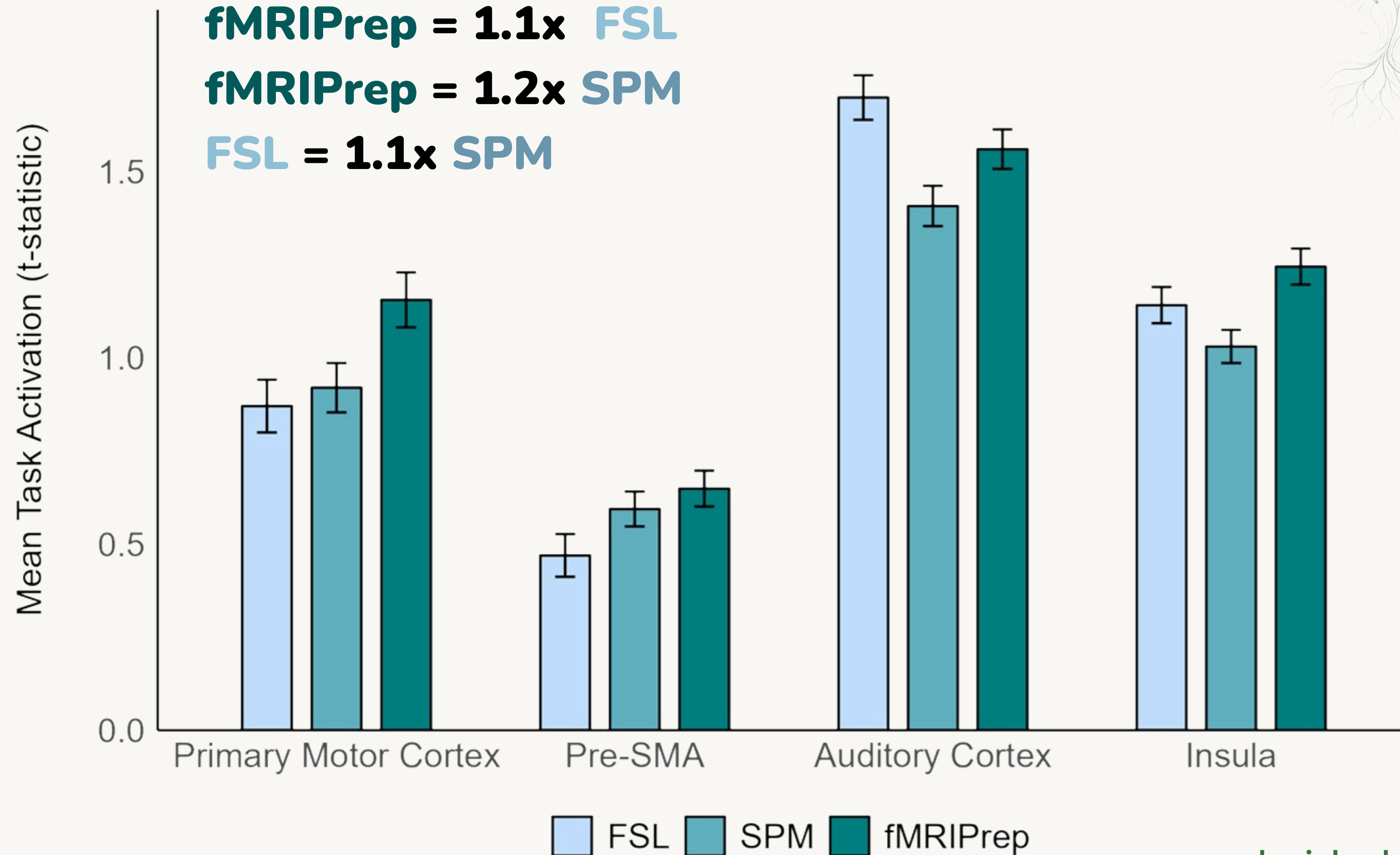


SPM = 1.3x FSL

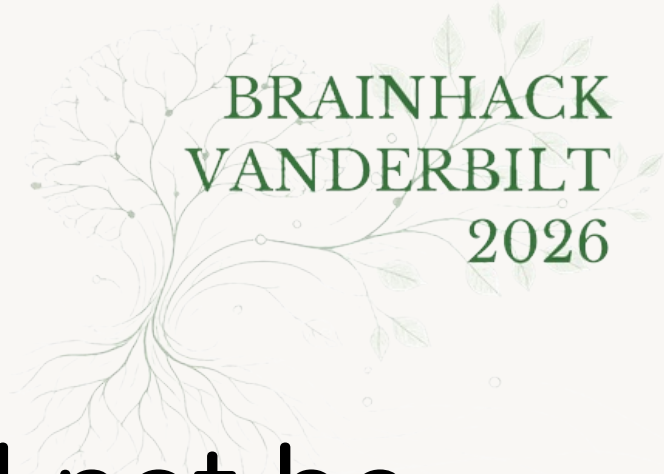
fMRIPrep = 30x FSL

fMRIPrep = 23x SPM

Statistical Sensitivity by Package

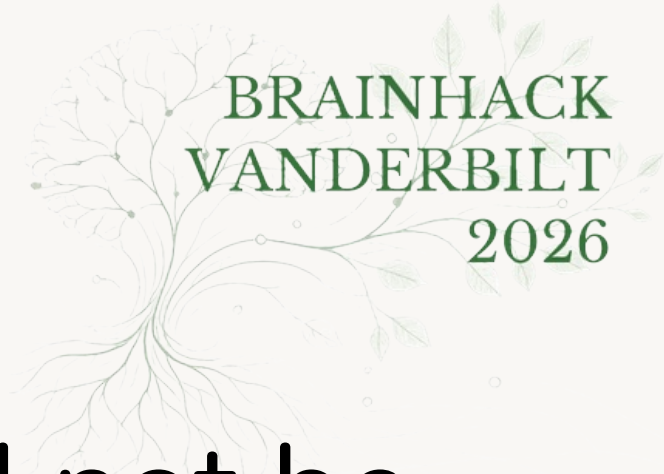


Recommendations



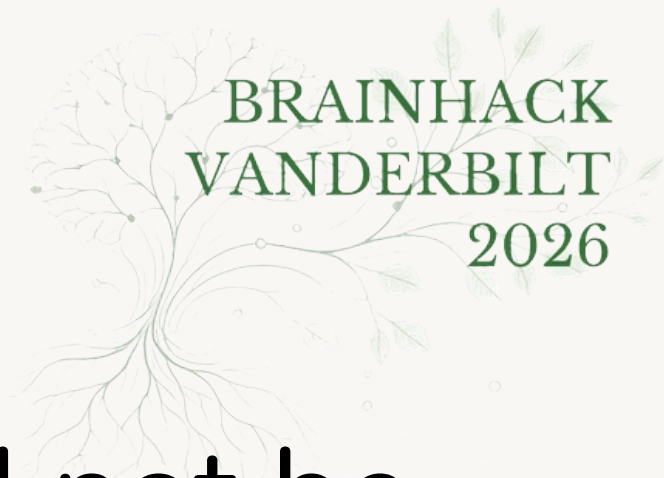
- Use of **high-compute tools** (like fMRIPrep) should not be demonised if they provide scientific gains

Recommendations



- Use of **high-compute tools** (like fMRIPrep) should not be demonised if they provide scientific gains
- When looking for robust effects, **low-compute tools** (FSL, SPM) may be sufficient

Recommendations



- Use of **high-compute tools** (like fMRIPrep) should not be demonised if they provide scientific gains
- When looking for robust effects, **low-compute tools** (FSL, SPM) may be sufficient
- **Tool developers** should work to reduce energy usage of preprocessing, providing ‘carbon-minimal’ flags. We need a balance between sustainability, performance, and usability

Ten recommendations

BOX 1. SUMMARY OF TEN RECOMMENDATIONS FOR REDUCING THE CARBON FOOTPRINT OF NEUROIMAGING COMPUTING

1. *Preregister a study analysis plan in order to avoid repetitions*
2. *Quantify and report the carbon footprint of your computing using available carbon tracking tools*
3. *Only run the preprocessing and analysis steps that you need*
4. *Run your computing at lower carbon intensity times and in lower carbon intensity locations*
5. *Regularly remove files that you do not need*
6. *Plan where, and for how long, you will store files, aided by research technicians*
7. *Advocate for non-commercial and centralised data storage solutions*
8. *Publicly share sufficient data to ensure it is FAIR (Findable, Accessible, Interoperable, Reusable), but consider the extent of what others will actually need or use*
9. *Make use of existing preprocessed data when possible, instead of acquiring and processing new data*
10. *Discuss the importance of greener computing with other neuroimagers and advocate for systemic change*



Souter et al. (2023)

The future of computing



AI is exhausting the power grid. Tech firms are seeking a miracle solution.

Some data centers need as much energy as a small city, turning companies that promised a clean energy future into some ...

[washingtonpost.com](https://www.washingtonpost.com)

Adoption of Machine Learning

Ongoing advances in **machine learning** and **artificial intelligence** will mean we use more energy for tasks where these methods wouldn't have previously been used. Will include energy usage for **training** and facilitating **tool use**.



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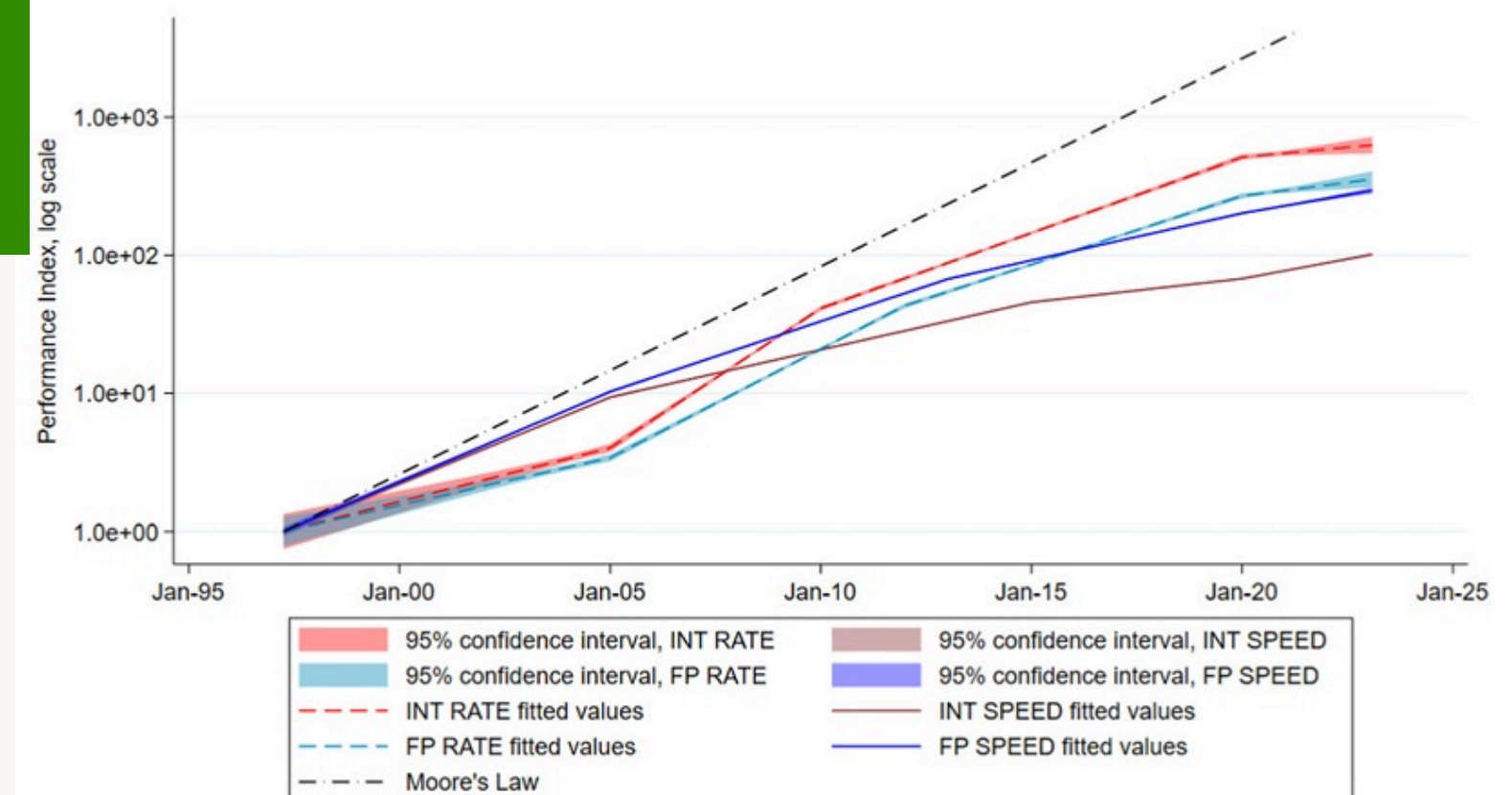
washingtonpost.com

Upgrades in Hardware

As hardware becomes more **efficient**, we can do **more work** for **less energy** (and therefore less carbon). We've seen good progress in recent decades, and this may continue.

Adoption of Machine Learning

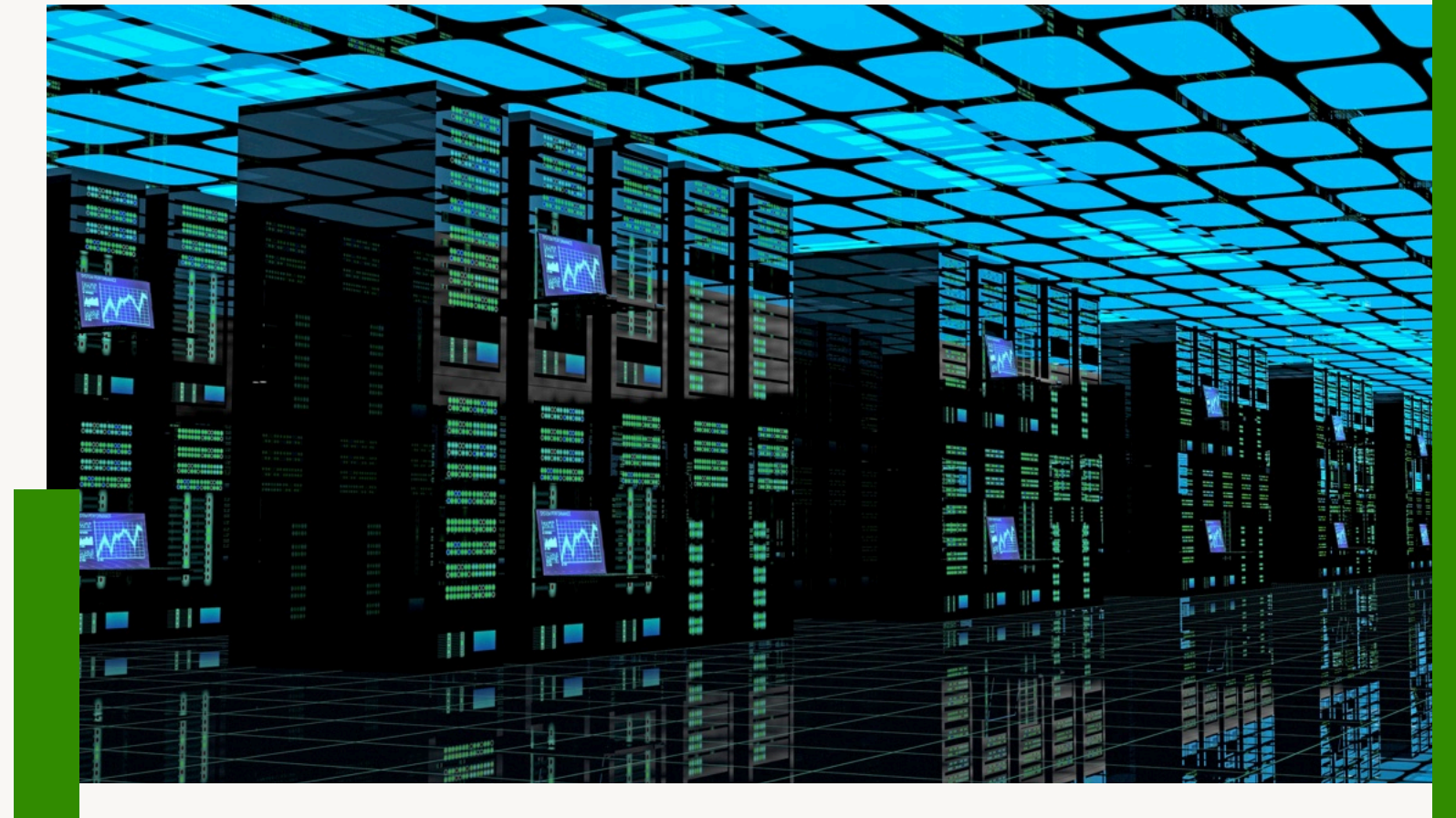
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Coyle & Hampton (2024),
Telecommunications Policy

Increased Centralisation

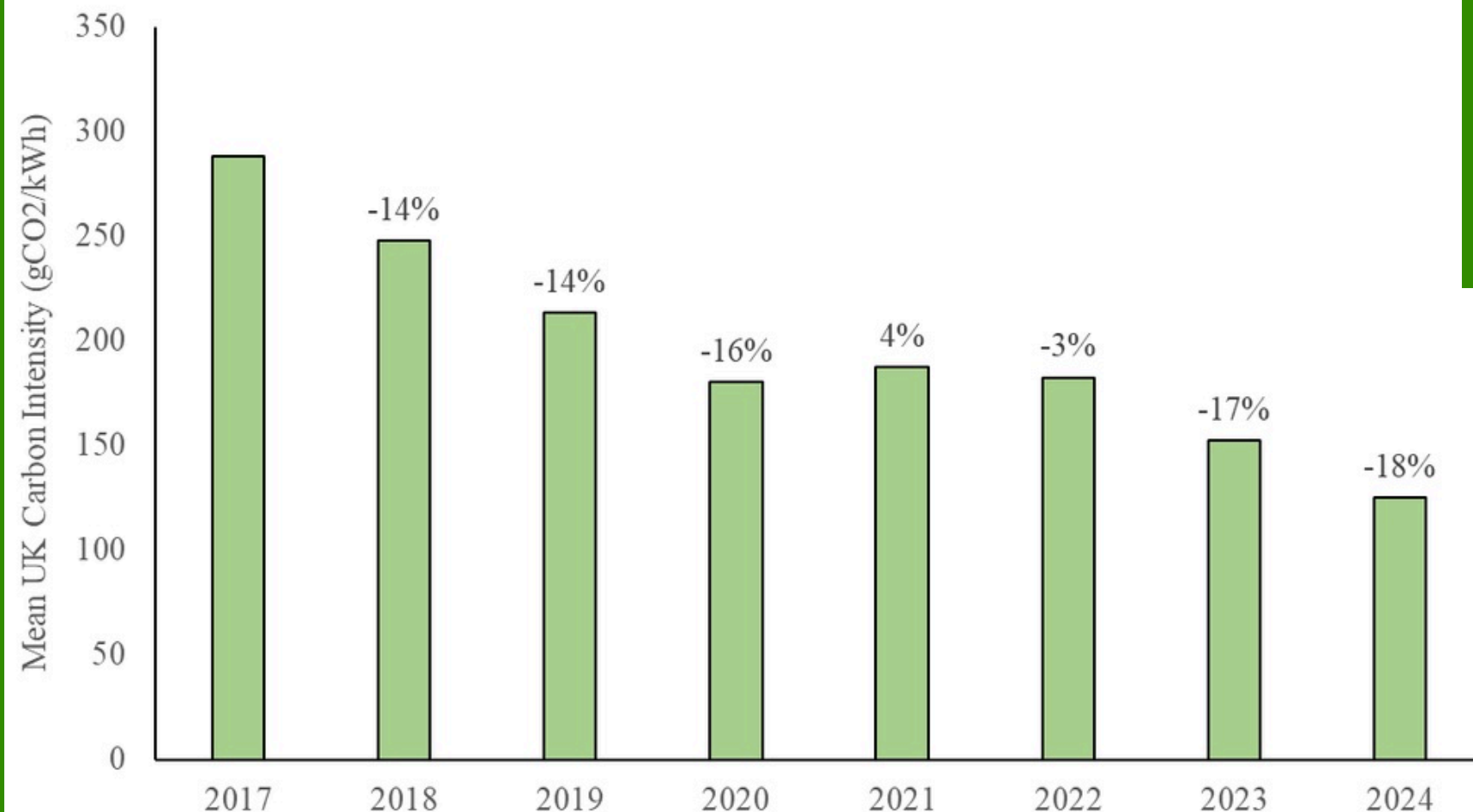
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Data from carbonintensity.org.uk

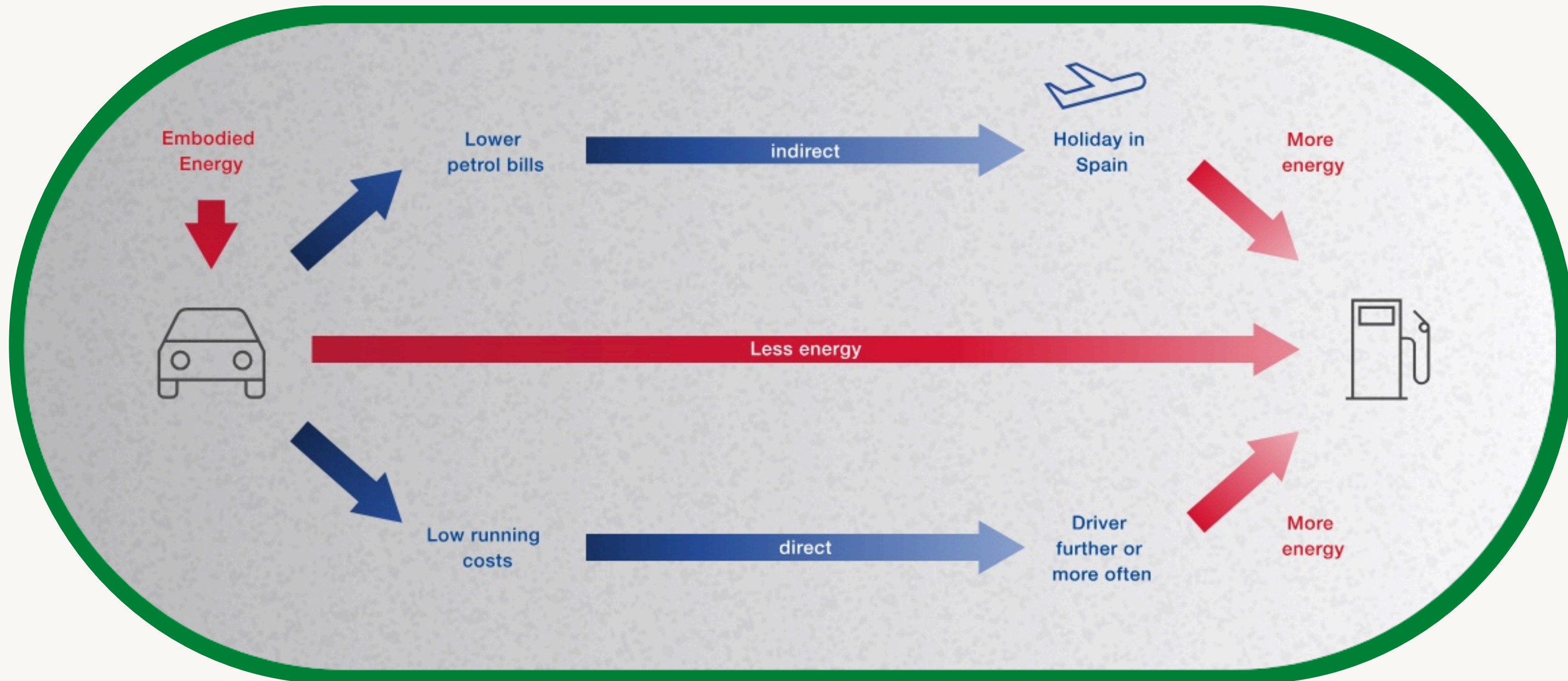


Decarbonisation of Energy

As the **carbon intensity** of our energy supplies decreases, we can use the same amount of energy for **less carbon**. However, infrastructure for renewable energy is limited.

Rebound Effects

Increasing cheap and efficient computing means we will likely use more of it, leading to a net increase in computing energy usage.



Source: Steven Sorrell: The Rebound Effect: an assessment of the evidence for economy-wide energy savings from improved energy efficiency; UK Energy Research Centre, London, 2007

The impact of AI



‘Just an unbelievable amount of pollution’: how big a threat is AI to the climate?

Defenders say AI can do good to fight the climate crisis. But spiralling energy and water costs leave experts worried

G the Guardian / Jan 6

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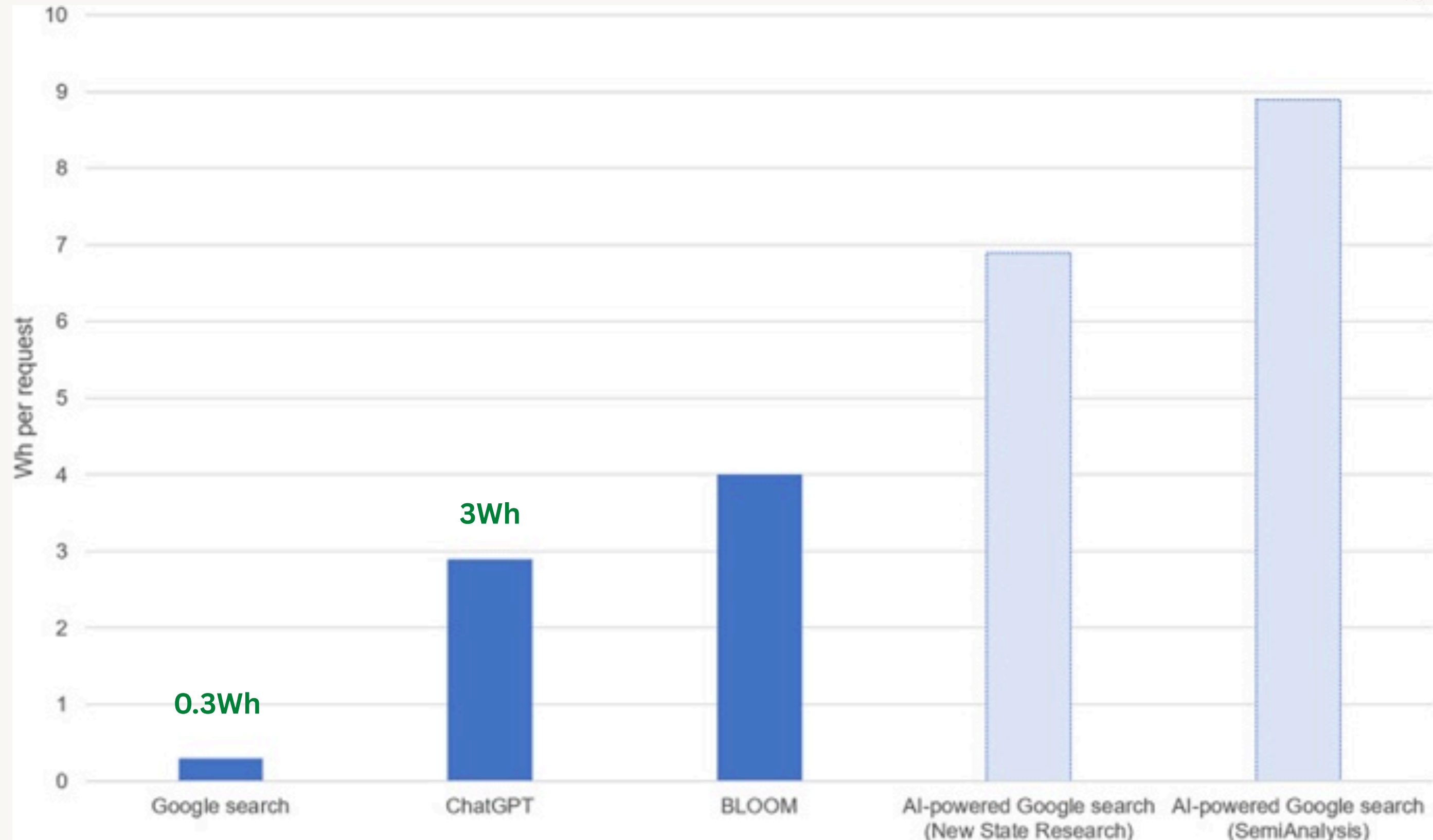
New study finds AI could reduce global emissions annually by 3.2 to 5.4 billion tonnes of carbon-dioxide-equivalent by 2035

AI advancements in power, transport and food consumption could reduce global emissions of greenhouse gases by 3.2 to 5.4 billion tonnes of carbon-dioxide-equivalent annually by 2035, according to new research.

LSE Grantham Research Institute on climate change and the environment /

Large Language Models

BRAINHACK
VANDERBILT
2026



de Vries (2023)

brainhack-vandy.github.io

Large Language Models



“It is probably too optimistic to expect that **improvements in hardware and software efficiencies** will fully offset any long-term changes in AI-related electricity consumption. These advancements can trigger a **rebound effect** whereby increasing efficiency leads to **increased demand for AI**, escalating rather than reducing total resource use”.

de Vries (2023)

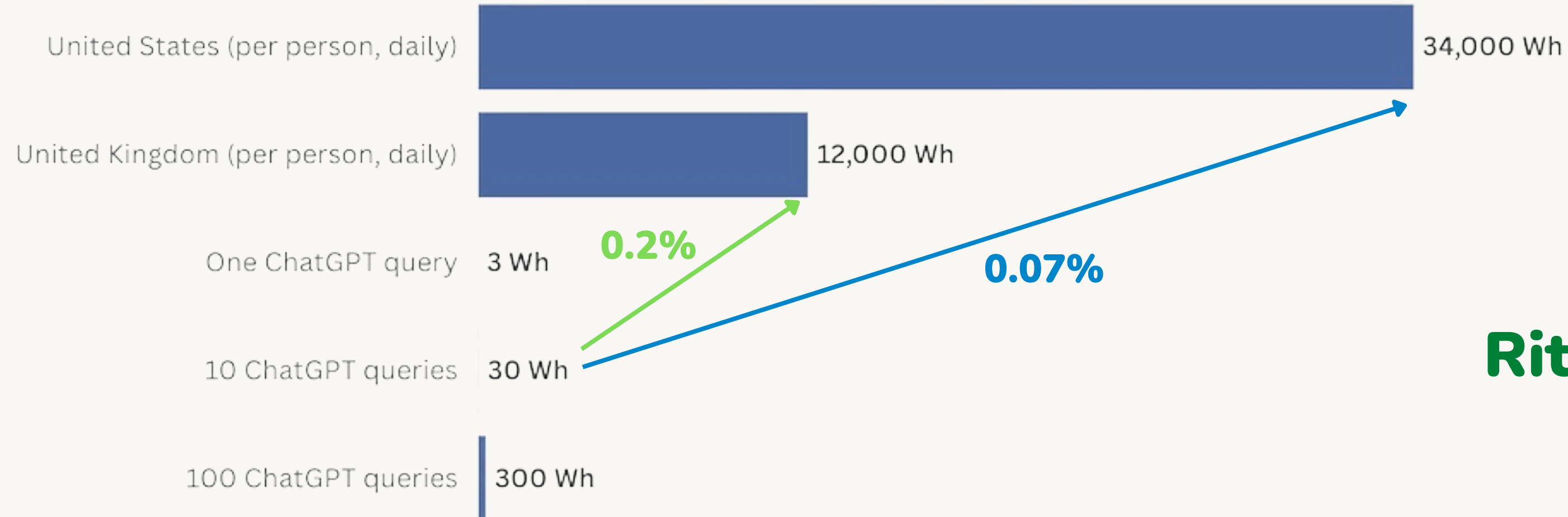


Large Language Models



How does the electricity use of ChatGPT compare to the daily average footprint in the UK and the US?

Measured in watt-hours. Electricity consumption for the US and UK includes household use plus services and industry within the economy.



Ritchie (2025)



Source: Ember Energy, Alex de Vries, and Andy Massley • This assumes that one ChatGPT query uses 3 watt-hours of electricity. More recent data suggests this could now be as low as 0.3 Wh.

Large Language Models



“More up-to-date analyses suggest that a ChatGPT query now uses just **0.3 Wh** — ten times less... That would mean that our already small environmental impact numbers from above are ten times too high. 10 queries a day would not be equal to **0.2%** of a Brit’s electricity consumption, but **0.02%** instead.”

Ritchie (2025)



Large Language Models



“We’re talking about whether our current calculations are **ten times too high**. Of course, tech companies do know what the right number is; it’s just that a **lack of transparency** means the rest of us are left bumbling around, wasting time.”

Ritchie (2025)



Large Language Models



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Large Language Models



“I am not saying that AI energy demand, on aggregate, is not a problem... It’s just that individuals querying chatbots is a **relatively small part** of AI's total energy consumption. That’s how both of these facts can be true at the same time.”

Ritchie (2025)



Green DiSC: a Digital Sustainability Certification

Green DiSC is the first open-access certification scheme which provides a roadmap for research groups and institutions who want to tackle the environmental impacts of their computing activities.

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THANK YOU!

FOR YOUR ATTENTION



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